

POLITECNICO
MILANO 1863

Monte Carlo Methods for the Boltzmann Equation: From DSMC to Time Relaxed Schemes

Report of:

Tommaso Fogarin

11173782

Advisor: Prof. Silvia Lorenzani
Course: Methods and Models for Statistical Mechanics
Academic Year 2025/2026

Contents

Contents **iii**

1 Introduction **1**

2 The Boltzmann equation **3**
2.1 The collision operator 4
2.2 Boundary conditions 4
2.3 The fluid-dynamic limit 5
2.4 Operator splitting 6

3 Fundamentals of Monte Carlo and Particle Methods **9**
3.1 What is a Monte Carlo method? 9
3.2 Monte Carlo integration and the law of large numbers 10
3.3 Statistical error: variance and the central limit theorem 10
3.4 Variance reduction: importance sampling 12
3.5 Sampling from a prescribed density 12
3.6 From integration to simulation: particle methods 13

4 DSMC and TRMC Particle Schemes **15**
4.1 Direct Simulation Monte Carlo (DSMC) 15
4.1.1 The collision rate of a particle system 16
4.1.2 Bird's scheme for Maxwell molecules 18
4.1.3 Bird's scheme for general VHS molecules 18
4.2 Time Relaxed Monte Carlo (TRMC) methods 20
4.2.1 The Wild sum and time relaxed schemes 20
4.2.2 First-order TRMC for VHS molecules 24
4.2.3 Higher-order TRMC schemes 25
4.2.4 Recursive TRMC methods 26
4.3 Generalization to the space inhomogeneous case 30
4.3.1 The splitting particle method 30
4.3.2 Implementation of boundary conditions 31

Bibliography **33**

A Appendix A **35**
A.1 Problem Setting . . . 35
A.2 Free Transport and the Gain–Loss Structure . . . 36
A.3 Kinematics of an Elastic Binary Collision between Hard Spheres . . . 36
A.4 Counting Collisions: The Two-Particle Distribution . . . 37
A.5 The Boltzmann–Grad Limit . . . 38
A.6 The Molecular Chaos Assumption and Closure . . . 39
A.7 General Collision Kernels . . . 39
A.8 The Knudsen Number and the Scaled Boltzmann Equation . . . 40

B Appendix B **43**
B.1 Univariate and discrete distributions . . . 43
B.1.1 Inversion . . . 43
B.1.2 Acceptance-rejection . . . 44
B.1.3 Composition . . . 44
B.1.4 Discrete distributions . . . 45
B.1.5 Sampling without repetition . . . 45
B.2 Multivariate distributions . . . 46
B.2.1 The factorised case . . . 46
B.2.2 The general case: sequential conditioning . . . 46
B.2.3 Multivariate acceptance-rejection . . . 47

List of Figures **49**

List of Tables **51**

1 | Introduction

This report presents an accessible review of Monte Carlo methods for the Boltzmann equation, based on the paper by Pareschi and Russo, *An Introduction to Monte Carlo Method for the Boltzmann Equation* [3]. The aim of this work is to go pedagogically through everything that is needed to fully comprehend the Time Relaxed Monte Carlo (TRMC) schemes introduced by Pareschi and Russo, and then to analyze them theoretically.

Beginning with kinetic theory, Chapter 2 introduces the Boltzmann equation, its collision operator, and the Variable Hard Sphere (VHS) collision model. It outlines the relevant boundary conditions, the operator splitting strategy used throughout particle methods, and the fluid-dynamic limit ($\epsilon \to 0$) that recovers the compressible Euler equations. Chapter 3 then builds the probabilistic framework required to construct these particle schemes. Starting from first principles, it covers integration, the law of large numbers, statistical error, and variance reduction techniques, before addressing general sampling algorithms. The chapter concludes by bridging the gap between static Monte Carlo integration and dynamic particle simulation. Chapter 4 focuses on the implementation and analysis of particle algorithms. It begins with Bird's classical Direct Simulation Monte Carlo (DSMC) scheme for both Maxwell and general VHS molecules. Next, it introduces the Wild sum representation of the space-homogeneous Boltzmann equation, establishing the theoretical derivation for first-order, higher-order, and recursive TRMC schemes. Particular attention goes to their Asymptotic Preserving (AP) property, which maintains numerical efficiency and accuracy uniformly across all scales of ϵ. Finally, the chapter extends these methods to space-inhomogeneous problems by combining operator splitting with physical boundary conditions. For interested readers, appendix A provides a self-contained, step-by-step heuristic derivation of the Boltzmann equation from the dynamics of N colliding hard spheres, making explicit the physical hypotheses that particle schemes themselves use; Appendix B, instead, is needed to understand in detail the general-purpose sampling techniques used throughout the report.

To ensure structural consistency, the report follows the logical progression laid out by Pareschi and Russo; however, I have expanded their framework wherever I felt it was necessary, adding my own derivations and personal perspective on the subject.

2 | The Boltzmann equation

In this first chapter, we focus on establishing the primary concepts and notation necessary to ensure a common baseline for the rest of the work. For readers seeking a deeper dive, Appendix A provides a self-contained, step-by-step heuristic derivation of the governing equations central to this report, explicitly detailing each physical assumption and its underlying intuition.

Let $f = f(x, v, t) \geq 0$ denote the one-particle distribution function of a dilute gas, where $x \in \Omega \subset \mathbb{R}^3$ is the physical position, $v \in \mathbb{R}^3$ is the microscopic velocity of a molecule, and $t > 0$ is time. The quantity $f(x, v, t)\,\mathrm{d}x\,\mathrm{d}v$ represents (up to a normalization by the molecular mass) the expected number of particles occupying the infinitesimal volume $\mathrm{d}x\,\mathrm{d}v$ of phase space around (x, v) at time t.

The evolution of f is governed by the Boltzmann equation

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f \;=\; \frac{1}{\varepsilon}\, Q(f, f), \tag{2.1}$$

and we consider the initial condition

$$f(x, v, t = 0) = f_0(x, v). \tag{2.2}$$

The left-hand side of (2.1) is the *free-streaming* (or transport) term: it describes the motion of particles that travel in absence of external forces and collisions in straight lines with constant velocity v. Instead, the right-hand side accounts for the effect of *binary collisions* between molecules, through the bilinear collision operator $Q(f, f)$.

The parameter $\varepsilon > 0$ is the *Knudsen number*, defined as the ratio between the mean free path of the molecules and a characteristic macroscopic length scale of the problem. Small values of ε correspond to a gas close to local thermodynamic equilibrium (the fluid regime), whereas values of order one or larger correspond to a strongly rarefied, genuinely kinetic regime. As will be discussed in Section 2.3, this single parameter controls the transition between the kinetic and the fluid-dynamic descriptions, and it is also the source of the main numerical difficulties encountered by particle methods, since the collision operator becomes stiffer and stiffer as $\varepsilon \to 0$.

2.1. The collision operator

The collision operator describes how the velocities of two colliding molecules are redistributed by an elastic binary collision. It reads

$$Q(f,f)(v) = \int_{\mathbb{R}^3}\int_{S^2} \left[f(v')f(v'_*) - f(v)f(v_*)\right] \sigma(\omega, |v-v_*|)\,\mathrm{d}\omega\,\mathrm{d}v_*, \tag{2.3}$$

where ω is a unit vector on the sphere S^2. The pre-collisional velocity of the two particles (v,v_*) and post-collisional velocities (v',v'_*) are linked, for an elastic and mechanically symmetric collision, by the relations

$$v' = \tfrac{1}{2}\big(v + v_* + |v-v_*|\,\omega\big), \qquad v'_* = \tfrac{1}{2}\big(v + v_* - |v-v_*|\,\omega\big), \tag{2.4}$$

which guarantee conservation of momentum and kinetic energy at each individual collision, $v + v_* = v' + v'_*$ and $|v|^2 + |v_*|^2 = |v'|^2 + |v'_*|^2$.

The nonnegative function σ, called the *scattering kernel* (or differential cross-section), encodes the microscopic physics of the interaction between molecules. For an interaction potential corresponding to an inverse k-th power force law, the kernel takes the form

$$\sigma(|v-v_*|,\theta) = b_\alpha(\theta)\,|v-v_*|^\alpha, \qquad \alpha = \frac{k-5}{k-1}. \tag{2.5}$$

A widely used numerical model is the *Variable Hard Sphere* (VHS) model, obtained by taking $b_\alpha(\theta) = C_\alpha$ constant. Two limiting cases are particularly relevant for both theory and computation:

- $\alpha = 0$: the kernel is independent of the relative velocity, corresponding to *Maxwell molecules*. This case is analytically the most tractable and admits explicit solutions in special situations.
- $\alpha = 1$: this yields the *Hard Sphere* gas, the classical model for elastic billiard-ball collisions also shown in Appendix A, and the most common choice in practical DSMC simulations.

2.2. Boundary conditions

Equations (2.1)–(2.2) must be complemented with boundary conditions whenever the physical domain Ω is bounded. If n denotes the inward unit normal to $\partial\Omega$, the incoming flux of particles ($v\cdot n > 0$) at the boundary is typically related to the outgoing flux ($v_*\cdot n < 0$) through a *reflective* condition of the form

$$|v\cdot n|\,f(x,v,t) = \int_{v_*\cdot n<0} |v_*\cdot n(x)|\,K(v_*\to v,x,t)\,f(x,v_*,t)\,\mathrm{d}v_*, \tag{2.6}$$

where the boundary kernel K is nonnegative and normalized so that mass is conserved at the wall,

$$K(v_*\to v,x,t) \ge 0, \qquad \int_{v\cdot n(x)\ge 0} K(v_*\to v,x,t)\,\mathrm{d}v = 1. \tag{2.7}$$

A particularly important instance is given by *Maxwell's boundary conditions*, which model a solid wall at which a fraction $\alpha \in [0,1]$ of the impinging molecules is absorbed and thermally re-emitted, while the remaining fraction $1-\alpha$ undergoes specular reflection. This is written as a convex combination of a specular reflection operator R and a re-emission (thermalization) operator M,

$$f(x,v,t) = (1-\alpha)\, Rf(x,v,t) + \alpha\, Mf(x,v,t), \qquad v\cdot n(x) \geq 0, \tag{2.8}$$

with

$$Rf(x,v,t) = f\big(x,\, v-2n(n\cdot v),\, t\big), \qquad Mf(x,v,t) = \mu(x,t)\, M_w(v), \tag{2.9}$$

where $M_w(v) = \exp(-|v|^2/2T_w)$ is the (unnormalized) Maxwellian corresponding to the wall temperature T_w, and $\mu(x,t)$ is obtained by imposing conservation of mass at the wall. The coefficient α is called the *accommodation coefficient*: for $\alpha = 0$ (pure specular reflection) the re-emitted molecules retain the mass, tangential momentum and temperature flux of the incoming ones, while for $\alpha = 1$ (full accommodation) they lose all memory of the incoming distribution except for the conservation of the number of particles, thus they completely accommodate the boundary wall.

A third, simpler type of boundary condition is the *inflow* condition, used for instance when computing stationary shock profiles: the distribution of the particles entering the domain is prescribed directly,

$$f(x,v,t) = g(v,t), \qquad x\in\partial\Omega,\ v\cdot n > 0, \tag{2.10}$$

typically taken to be a Maxwellian.

2.3. The fluid-dynamic limit

The Boltzmann collision operator possesses two structural properties that are essential both for the physical consistency of the model and for the construction of numerical schemes.

Conservation. The collision operator conserves mass, momentum, and kinetic energy pointwise in (x,t):

$$\int_{\mathbb{R}^3} Q(f,f)\,\phi(v)\,\mathrm{d}v = 0, \qquad \phi(v) = 1,\ v,\ |v|^2. \tag{2.11}$$

The functions $\phi(v) = 1, v, |v|^2$ are called the *collision invariants*.

The H-theorem. The collision operator satisfies Boltzmann's celebrated H-theorem,

$$\int_{\mathbb{R}^3} Q(f,f)\,\log f\,\mathrm{d}v \ \leq\ 0. \tag{2.12}$$

This inequality is the kinetic counterpart of the second law of thermodynamics: it guarantees that the entropy of the gas can only increase (or stay constant) in time. As a

consequence, any equilibrium distribution, i.e. any f for which $Q(f,f)=0$, must be a *local Maxwellian*

$$M(\rho,u,T)(v)=\frac{\rho}{(2\pi T)^{3/2}}\exp\left(-\frac{|u-v|^2}{2T}\right), \tag{2.13}$$

where the density ρ, mean velocity u, and temperature T are the moments of f,

$$\rho=\int_{\mathbb{R}^3} f\,\mathrm{d}v, \qquad u=\frac{1}{\rho}\int_{\mathbb{R}^3} v\,f\,\mathrm{d}v, \qquad T=\frac{1}{3\rho}\int_{\mathbb{R}^3}|v-u|^2 f\,\mathrm{d}v. \tag{2.14}$$

As the Knudsen number $\varepsilon\to 0$, the collision term on the right-hand side of (2.1) forces the distribution function to relax, on a time scale much faster than the transport time scale, toward the local Maxwellian (2.13). Taking the first three velocity moments ($\phi=1, v, |v|^2$) of the Boltzmann equation, and using (2.11) together with the fact that $f\to M(\rho,u,T)$, one obtains, to leading order in ε, the closed system of the compressible Euler equations of gas dynamics,

$$\begin{cases} \dfrac{\partial\rho}{\partial t}+\nabla_x\cdot(\rho u)=0,\\[2mm] \dfrac{\partial(\rho u)}{\partial t}+\nabla_x\cdot(\rho u\otimes u+p\,\mathbb{I})=0, \qquad p=\rho T,\quad E=\tfrac{3}{2}\rho T+\tfrac{1}{2}\rho|u|^2,\\[2mm] \dfrac{\partial E}{\partial t}+\nabla_x\cdot\big((E+p)u\big)=0, \end{cases} \tag{2.15}$$

where p is the pressure and E the total energy density.

This limit is the key motivation for the whole methodology discussed in this report: standard particle schemes for the Boltzmann equation become extremely expensive precisely in the regime $\varepsilon\ll 1$ where (2.15) becomes valid, because the average time between collisions vanishes as $\varepsilon\to 0$ and an enormous number of collisions per unit time must be resolved. This observation motivates the search for numerical schemes, the Time Relaxed Monte Carlo (TRMC) methods discussed in Chapter 4, that remain efficient uniformly in ε, and that are *asymptotic preserving*, i.e. that automatically degenerate into a consistent scheme for the limiting Euler equations as $\varepsilon\to 0$.

2.4. Operator splitting

From the numerical point of view, the most common strategy to solve the full Boltzmann equation (2.1)–(2.2) is to decouple the transport and the collision mechanisms by means of an *operator splitting*. Over a time step Δt, the solution is advanced in two stages.

1. **Collision step.** One first solves, for every $x\in\Omega$, the space-homogeneous problem

$$\begin{cases} \dfrac{\partial\bar f}{\partial t}=\dfrac{1}{\varepsilon}\,Q(\bar f,\bar f),\\[2mm] \bar f(x,v,0)=f_0(x,v), \end{cases} \tag{2.16}$$

   over the time interval $[0,\Delta t]$, where $\bar f(x,v,\tau)$ denotes the intermediate stage variable representing the velocity-space distribution evolution at a fixed spatial position x.

2. **Transport (convection) step.** The result of the collision step is then used as the initial datum for the free-streaming equation

$$\begin{cases} \dfrac{\partial f}{\partial t} + v \cdot \nabla_x f = 0, \\ f(x, v, 0) = \bar{f}(x, v, \Delta t), \end{cases} \tag{2.17}$$

   again integrated over $[0, \Delta t]$.

Iterating this two-stage procedure produces an approximation of the solution at the successive discrete times $n\Delta t$. The advantage of this splitting is that it isolates, in the collision step (2.16), essentially all of the numerical difficulty of the problem: the transport step (2.17) is a simple linear advection equation, while the nonlinear collision operator becomes increasingly stiff as the fluid regime $\varepsilon \ll 1$ is approached. This is exactly the splitting used, together with a particle discretization of the collision step, by the direct simulation Monte Carlo (DSMC) and time relaxed Monte Carlo (TRMC) methods discussed in chapter 4. The simple splitting described above is only first-order accurate in time; higher-order splittings (e.g. Strang splitting) exist, but for particle methods it is not clear whether increasing the order of the splitting is beneficial, since the stochastic error typically introduced by the Monte Carlo discretization of the collision step is itself only of low order.

3 | Fundamentals of Monte Carlo and Particle Methods

Before introducing the numerical algorithms used to solve the Boltzmann equation studied in the previous chapter, it is essential to establish the mathematical foundations on which they rely. To this end, this chapter provides a self-contained and pedagogical overview of the general Monte Carlo framework.

3.1. What is a Monte Carlo method?

A *Monte Carlo method* is, broadly speaking, any numerical technique that uses sequences of random numbers to compute a deterministic quantity, or to simulate the evolution of a system whose exact dynamics is too complex to be integrated directly. The name, coined by Metropolis and Ulam in the 1940s after the Monte Carlo casino in Monaco, refers to the element of chance that drives the algorithm, even though the quantity being computed, e.g. an integral, a moment, the solution of a PDE, is entirely deterministic. The first large-scale application of the method, carried out by Ulam, von Neumann and Metropolis at Los Alamos, was the simulation of neutron transport, i.e. of a *linear* kinetic equation structurally very close to the Boltzmann equation. The particle methods discussed in this chapter can be seen as the extension of that same idea to the genuinely nonlinear collision operator.

Throughout this section we assume that the computer has at its disposal an inexhaustible sequence $\xi_1, \xi_2, \ldots$ of independent random numbers, each uniformly distributed on $[0, 1]$. In practice these are produced by a deterministic pseudo-random number generator with extremely long period; the construction of such generators is a non trivial subject in its own right and lies outside the scope of this report (see [5] for an introduction). What matters for us is that *any* other random variable we need (a velocity sampled from a Maxwellian, a deflection angle sampled from a scattering kernel, a collision time) must ultimately be constructed, by one of the techniques that will be shown below and investigated more carefully in Appendix B, out of such uniform random numbers.

3.2. Monte Carlo integration and the law of large numbers

The prototypical Monte Carlo problem is the computation of an integral. Let $D \subseteq \mathbb{R}^d$, let p be a probability density on D (i.e. $p \geq 0$ and $\int_D p(x)\,dx = 1$), and let $\phi : D \to \mathbb{R}$ be a function that is square-integrable with respect to p. We wish to compute

$$I \;=\; \int_D \phi(x)\,p(x)\,dx \;=\; \mathbb{E}_p[\phi(X)], \tag{3.1}$$

i.e. the expectation of ϕ under a random variable X with law p. (Ordinary integration over a domain D of finite measure is recovered by taking p to be the uniform density on D; the general form above is the one that will be useful later, when p will be replaced by the collision kernel σ or by a local Maxwellian.)

Given independent samples $X_1, \dots, X_N$ drawn from p, the *Monte Carlo estimator* of I is the empirical average

$$\widehat{I}_N \;=\; \frac{1}{N}\sum_{n=1}^{N} \phi(X_n). \tag{3.2}$$

Two elementary facts justify the use of (3.2).

Unbiasedness. Since the X_n are identically distributed with law p,

$$\mathbb{E}\big[\widehat{I}_N\big] = \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}[\phi(X_n)] = \mathbb{E}_p[\phi(X)] = I \qquad \text{for every } N \geq 1.$$

Consistency (strong law of large numbers). If $\mathbb{E}_p|\phi(X)| < \infty$, Kolmogorov's strong law of large numbers guarantees that

$$\widehat{I}_N \;\xrightarrow[N\to\infty]{\text{a.s.}}\; I. \tag{3.3}$$

Together, unbiasedness and (3.3) say that $\widehat{I}_N$ is always centered at the right value and converges to it almost surely as more samples are used, with no assumption whatsoever on the smoothness of ϕ, on the regularity of D, or, crucially, on the dimension d. This last point is the entire reason why Monte Carlo methods, and not deterministic quadrature rules, are the tool of choice for the Boltzmann equation, as we will see later.

3.3. Statistical error: variance and the central limit theorem

The law of large numbers guarantees convergence but says nothing about the *rate* (speed of convergence). To quantify the error at finite N we introduce the variance of ϕ under p,

$$\sigma^2(\phi) \;=\; \mathrm{Var}_p\big(\phi(X)\big) \;=\; \int_D \big(\phi(x) - I\big)^2\,p(x)\,dx \;=\; \mathbb{E}_p[\phi(X)^2] - I^2, \tag{3.4}$$

which we assume finite. By independence of the X_n,

$$\mathrm{Var}\left(\widehat{I}_N\right) = \frac{1}{N^2}\sum_{n=1}^{N}\mathrm{Var}\left(\phi(X_n)\right) = \frac{\sigma^2(\phi)}{N}, \tag{3.5}$$

so that the *root-mean-square* error, or *statistical error*, of the Monte Carlo estimator is

$$\left(\mathbb{E}\left[(\widehat{I}_N - I)^2\right]\right)^{1/2} = \frac{\sigma(\phi)}{\sqrt{N}}. \tag{3.6}$$

A first, non-asymptotic, control on the fluctuations follows immediately from Chebyshev's inequality applied to (3.5): for every $k > 0$,

$$\mathbb{P}\left(\left|\widehat{I}_N - I\right| \geq k\,\frac{\sigma(\phi)}{\sqrt{N}}\right) \;\leq\; \frac{1}{k^2}. \tag{3.7}$$

A sharper, asymptotic statement is provided by the central limit theorem: if $\sigma(\phi) > 0$,

$$\frac{\sqrt{N}\left(\widehat{I}_N - I\right)}{\sigma(\phi)} \;\xrightarrow[N\to\infty]{d}\; \mathcal{N}(0,1), \tag{3.8}$$

so that, for N large, $\widehat{I}_N$ is approximately normal with mean I and standard deviation $\sigma(\phi)/\sqrt{N}$; this is what makes it customary to quote a Monte Carlo result as $\widehat{I}_N \pm \sigma(\phi)/\sqrt{N}$, with the understanding that the true value lies in this interval with probability $\approx 68\%$, or in $\widehat{I}_N \pm 2\sigma(\phi)/\sqrt{N}$ with probability $\approx 95\%$. In practice $\sigma^2(\phi)$ is itself unknown and is replaced by the unbiased sample variance

$$S_N^2 \;=\; \frac{1}{N-1}\sum_{n=1}^{N}\left(\phi(X_n) - \widehat{I}_N\right)^2. \tag{3.9}$$

A very important aspect is that (3.6) has a convergence rate $\mathcal{O}(N^{-1/2})$ strictly *independent* of the dimension d of D. In contrast, standard deterministic quadrature rules suffer from the "curse of dimensionality": tensorising a one-dimensional rule of order q across d dimensions requires $N = n^d$ evaluations and yields an error scaling as $\mathcal{O}(N^{-q/d})$. As d grows, the deterministic convergence rate deteriorates rapidly and is quickly overtaken by the dimension-free Monte Carlo rate.

This advantage is fundamental when dealing with the Boltzmann collision operator (2.3). Evaluating $Q(f,f)(v)$ at a single point requires a 5-dimensional integration over $(v_*, \omega) \in \mathbb{R}^3 \times S^2$. If one attempts to evaluate this integral deterministically at every node of a discretized 3D velocity grid, the total computational cost explodes as $O(N_v^3 \cdot N_v^3 \cdot N_\omega) = O(N_v^6 N_\omega)$ operations per time step, where N_v and N_ω represent the grid resolutions in velocity space and on the sphere, respectively. This number becomes intractable already for a modest N_v of a few tens of points per direction. It is exactly this combinatorial explosion that motivates replacing f with a finite set of numerical particles, and the collision integral with a Monte Carlo, rather than deterministic, evaluation: a particle scheme, as we shall see, updates N_p particles at a cost of order N_p (or $N_p \log N_p$) per time step, independently of the dimension of velocity space.

3.4. Variance reduction: importance sampling

Because the statistical error (3.6) decreases only like $N^{-1/2}$, it is often worthwhile to reduce the *prefactor* $\sigma(\phi)$ rather than increase N. The most important such technique, and the one most relevant to particle methods for the Boltzmann equation, is *importance sampling*. Suppose we want to compute $I = \int_D \phi(x)\,dx$ but sampling directly from a density proportional to ϕ can be impractical. Let q be *any* probability density on D with $q(x) > 0$ wherever $\phi(x) \neq 0$. Then

$$I = \int_D \frac{\phi(x)}{q(x)}\, q(x)\,dx = \mathbb{E}_q\left[\frac{\phi(X)}{q(X)}\right], \tag{3.10}$$

which suggests the estimator

$$\widehat{I}_N = \frac{1}{N}\sum_{n=1}^{N} \frac{\phi(X_n)}{q(X_n)}, \qquad X_n \overset{\text{iid}}{\sim} q, \tag{3.11}$$

still unbiased by the same argument as before, but now with variance $\sigma^2(\phi/q)/N$ governed by the new integrand ϕ/q. Choosing q proportional to $|\phi|$ makes the ratio ϕ/q (nearly) constant and drives the variance towards zero; of course $q \propto \phi$ can be constructed exactly only if I is already known (since it is the normalization constant of the probability distribution), so in practice one simply chooses q to *resemble* ϕ in shape, subject to the constraint that one must be able to sample from q efficiently. One must also take care that q does not vanish, or become too small, on a set where ϕ does not: this can make $\sigma^2(\phi/q)$ arbitrarily large, or even infinite, and the resulting estimator unreliable even though it remains formally unbiased.

3.5. Sampling from a prescribed density

Both the initialisation of a particle scheme (drawing N_p velocities from the initial data f_0) and each collision event (drawing a post-collisional direction ω from the scattering kernel, or a candidate pair from a majorant of σ) require the ability to sample from a *prescribed* density p_x, given only a source of uniform random numbers $\xi \sim \mathcal{U}[0,1]$. Two general-purpose constructions are used throughout the rest of this chapter and are recalled here; a complete catalogue of sampling techniques, including the multivariate and composition methods needed for the specific kernels appearing in the DSMC and TRMC schemes, is developed in Appendix B, following the treatment of [3].

Inversion. If p_x is a density on $\mathbb{R}$ with cumulative distribution function $P_x(x) = \int_{-\infty}^{x} p_x(y)\,dy$, and if P_x can be inverted, then

$$X = P_x^{-1}(\xi), \qquad \xi \sim \mathcal{U}[0,1], \tag{3.12}$$

is distributed according to p_x: indeed P_x is non-decreasing, so $\mathbb{P}(X \le x) = \mathbb{P}(P_x^{-1}(\xi) \le x) = \mathbb{P}(\xi \le P_x(x)) = P_x(x)$, which is the definition of $X \sim p_x$.

Acceptance-rejection. When P_x^{-1} is not available in closed form, one can instead bound p_x by a majorant $w(x) \geq p_x(x)$ for every x, chosen so that w has an invertible primitive W and a finite total mass $A = \int w(x)\,dx$. Sampling then proceeds as follows:

1. draw a candidate X from the density w/A by inversion, solving $W(X) = A\,\xi_1$;
2. draw an independent $\xi_2 \sim \mathcal{U}[0,1]$; accept X if $\xi_2\, w(X) < p_x(X)$, otherwise reject X and return to step 1.

The accepted samples are exactly distributed according to p_x. Indeed, for any x,

$$\mathbb{P}\big(X \leq x,\ \text{accepted}\big) = \int_{-\infty}^{x} \frac{w(y)}{A} \cdot \frac{p_x(y)}{w(y)}\,dy = \frac{P_x(x)}{A},$$

so that $\mathbb{P}(\text{accepted}) = 1/A$ and, conditionally on acceptance, X has distribution function P_x. The acceptance probability $1/A$ measures the efficiency of the scheme: the closer w hugs p_x, the fewer candidates are wasted. We will find this geometric idea, accept or reject a candidate collision by comparing it against a majorant of the collision kernel, in the Bird's particle schemes, and it's the same idea we'll meet again, in velocity space, in a moment.

3.6. From integration to simulation: particle methods

The discussion so far has treated Monte Carlo as a device for estimating a *single, fixed* number, $I = \mathbb{E}_p[\phi(X)]$. To use it for the collision step (2.16) we must take one further, conceptual step: instead of a fixed density p, we consider the *unknown, time-dependent* density $\bar{f}(\cdot,t)/\rho$ (where $\rho = \int \bar{f}\,dv$ is the conserved number density) that we are trying to compute in the first place.

The idea, due in this form to Kac and at the root of every particle method for kinetic equations, is to represent $\bar{f}(\cdot,t)$ not by its values on a velocity grid, but by the positions of N_p numerical particles $v_1(t), \ldots, v_{N_p}(t) \in \mathbb{R}^3$, through the empirical measure

$$f_{N_p}(v,t) \;=\; \frac{\rho}{N_p} \sum_{i=1}^{N_p} \delta\big(v - v_i(t)\big). \tag{3.13}$$

For a fixed time t, and provided the $v_i(t)$ were independent samples from $\bar{f}(\cdot,t)/\rho$, *any macroscopic moment* $\langle\phi\rangle(t) = \int \phi(v)\,\bar{f}(v,t)\,dv$ *would simply be the Monte Carlo estimator* (3.2) *studied in this section* built out of the particles,

$$\int \phi(v)\, f_{N_p}(v,t)\,dv = \frac{\rho}{N_p} \sum_{i=1}^{N_p} \phi(v_i(t)) \;\approx\; \langle\phi\rangle(t),$$

with the dimension-independent accuracy of Section 3.3.

The genuinely new ingredient in DSMC and TRMC is that particles are not resampled from an explicitly reconstructed probability density at each time step. Instead, particles

collide directly with one another in pairs according to a stochastic rule, constructed via importance sampling and acceptance-rejection, derived from the collision cross-section σ. The state of the particle ensemble $(v_1(t), \ldots, v_{N_p}(t))$ thus evolves as a Markov jump process on $(\mathbb{R}^3)^{N_p}$. Consequently, DSMC and TRMC operate as Monte Carlo methods not in the static sense of high-dimensional quadrature (Section 3.2), but in the dynamic sense of *stochastic particle simulation*: it is the empirical measure (3.13) of the N_p-particle ensemble, rather than any individual velocity trajectory $v_i(t)$, that approximates the kinetic solution $\bar{f}(\cdot, t)$ of (2.16).

The mathematical justification for this approximation as $N_p \to \infty$ rests on the propagation of chaos property discussed in Appendix A. The molecular chaos hypothesis (A5), which decouples pre-collisional states to derive the Boltzmann equation from deterministic N-body dynamics, is algorithmically mirrored in the numerical scheme by the independent, random pairing of simulation particles at each collision step. In this sense, DSMC and TRMC do not merely discretize equation (2.16); they simulate a stochastic N_p-particle jump process whose mean-field limit as $N_p \to \infty$ is the Boltzmann equation. Thus, the law of large numbers underlying Monte Carlo convergence is, in a precise mathematical sense, the same asymptotic principle that recovers kinetic continuum dynamics from microscopic particle interactions.

4 | DSMC and TRMC Particle Schemes

We can now put these tools to work and build the DSMC and TRMC algorithms. By combining the operator-splitting framework for the Boltzmann equation introduced in Chapter 2 with the Monte Carlo sampling techniques and stochastic simulation principles developed in Chapter 3, we begin by formulating particle-based numerical schemes for the space-homogeneous collision step,

$$\frac{\partial \bar{f}}{\partial t} = \frac{1}{\varepsilon} Q(\bar{f}, \bar{f}), \qquad \bar{f}(x, v, 0) = f_0(x, v),$$

which, we recall, is solved independently at every spatial point $x \in \Omega$ and is then coupled to the free-transport step (2.17) by the splitting scheme of Section 2.4.

Accordingly, the first two sections of this chapter are entirely devoted to the velocity-space collision mechanics of DSMC and TRMC, focusing on how these schemes evaluate $Q(\bar{f}, \bar{f})$ to evolve $\bar{f}(v, t)$ in time. The complete numerical solution of the kinetic problem, which also includes the spatial particle advection during the free-streaming step, and the generalization to the full space-inhomogeneous problem is postponed to Section 4.3.

Because physical transport is decoupled during the collision step, the spatial coordinate x acts as a frozen parameter. Furthermore, by the fundamental conservation property (2.11), the local mass density $\rho = \int_{\mathbb{R}^3} \bar{f}(v, t)\, \mathrm{d}v$ remains strictly constant in time throughout the collision process. We may therefore suppress x from the notation without loss of generality across the first two sections of this chapter.

4.1. Direct Simulation Monte Carlo (DSMC)

Direct Simulation Monte Carlo, introduced by G. A. Bird in the 1960s [1] and only later connected rigorously to the Boltzmann equation, is the most direct way to simulates the particle system as a continuous-time Markov jump process, with no further approximation of the collision mechanism itself. In what follows, we first formalize the concept of *collision rate* underlying this formulation and then specialize it to Bird's scheme, starting with Maxwell molecules before extending the treatment to general Variable Hard Sphere (VHS) interactions.

4.1.1. The collision rate of a particle system

First of all we start by splitting the collision operator as:

$$Q(f,f)(v) = Q^+(f,f)(v) - f(v)\,\nu[f](v), \tag{4.1}$$

where

$$Q^+(f,f)(v) = \int_{\mathbb{R}^3}\int_{S^2} \sigma(\omega, |v-v_*|)\, f(v')f(v'_*)\, d\omega\, dv_*, \qquad \nu[f](v) = \int_{\mathbb{R}^3} \sigma_T(|v-v_*|)\, f(v_*)\, dv_*, \tag{4.2}$$

and

$$\sigma_T(r) := \int_{S^2} \sigma(\omega, r)\, d\omega \tag{4.3}$$

is the *total*, angle-integrated cross section. This decomposition is mathematically justified under angular cutoff or VHS assumptions, where the total cross section $\sigma_T(r) < \infty$ remains finite. The quantity $\nu[f](v)/\varepsilon$ carries a direct probabilistic interpretation: it represents the expected collision rate per unit macroscopic time for a particle with velocity v immersed in a background distribution f, with corresponding mean free time $\varepsilon/\nu[f](v)$. This frequency acts as the generator of the underlying jump process, and it is this quantity, rather than the full operator Q, that DSMC discretizes directly.

To discretize this continuous frequency within the particle framework introduced in Section 3.6, we postulate, consistently with the propagation-of-chaos heuristic invoked in Appendix A.6, that each unordered pair of particles $\{i,j\}$ $(i \neq j)$ collides independently with rate

$$\lambda_{ij}(t) := \frac{\rho}{\varepsilon\,(N_p-1)}\,\sigma_T\big(|v_i(t)-v_j(t)|\big). \tag{4.4}$$

The denominator $N_p - 1$ represents the number of candidate collision partners available to particle i. Fixing particle i and summing over all $j \neq i$, the Strong Law of Large Numbers guarantees that as $N_p \to \infty$, the sample average over the empirical particle distribution converges

$$\sum_{j\neq i}\lambda_{ij}(t) = \frac{\rho}{\varepsilon}\cdot\frac{1}{N_p-1}\sum_{j\neq i}\sigma_T\big(|v_i-v_j|\big) \xrightarrow[N_p\to\infty]{\text{a.s.}} \frac{\rho}{\varepsilon}\int_{\mathbb{R}^3}\sigma_T(|v_i-v_*|)\left(\frac{\bar f(v_*)}{\rho}\right)dv_* = \frac{\nu[\bar f](v_i)}{\varepsilon}, \tag{4.5}$$

where $\bar f(v_*)/\rho$ is the normalized particle velocity probability density (integrating to 1). The density factor ρ cancels cleanly upon integration, recovering the continuous collision frequency $\nu[\bar f](v_i)/\varepsilon$. Thus, the total rate at which a tagged particle collides with the rest of the N_p-particle ensemble converges, as $N_p \to \infty$, to the exact continuum collision frequency of (4.2). This limit provides the analogue of the correspondence between particle dynamics and kinetic theory previously established at the N-body level via the Boltzmann–Grad limit (Appendix A.5).

Summing the pairwise rate $\lambda_{ij}(t)$ over the $\binom{N_p}{2}$ possible unordered pairs gives the total rate at which *some* collision occurs in the system,

$$R(t) := \sum_{1\le i<j\le N_p} \lambda_{ij}(t) = \frac{\rho}{\varepsilon(N_p-1)} \sum_{i<j} \sigma_T\big(|v_i - v_j|\big) = \frac{N_p\,\rho\,\bar\sigma(t)}{2\varepsilon}, \tag{4.6}$$

where

$$\bar\sigma(t) := \binom{N_p}{2}^{-1} \sum_{1\le i<j\le N_p} \sigma_T\big(|v_i(t) - v_j(t)|\big) \tag{4.7}$$

is the empirical mean of the total cross section over all pairs currently present in the system. Freezing R over a short time step Δt (exactly as the collision step of the splitting scheme of Section 2.4 freezes x), the expected number of collisions during $[n\Delta t, (n+1)\Delta t]$ and the corresponding mean collision time are

$$N_c = R\,\Delta t = \frac{N_p\,\rho\,\bar\sigma\,\Delta t}{2\varepsilon}, \qquad \Delta t_c = \frac{1}{R} = \frac{2\varepsilon}{N_p\,\rho\,\bar\sigma}. \tag{4.8}$$

The time increment Δt_c defines the stochastic simulation clock driving the jump process in the DSMC algorithm.

A fundamental distinction arise when considering the scattering kernel for VHS molecules $b_\alpha(\theta) = C_\alpha$ which is independent of the scattering angle, so (4.3) evaluates explicitly to

$$\sigma_T(r) = 4\pi C_\alpha\, r^\alpha, \tag{4.9}$$

and, because $b_\alpha(\theta)$ is angle-independent, the post-collisional direction ω is *always* uniformly distributed on S^2, whatever the value of α. However, the total cross section (4.9), and hence the collision *rate*, depends on α and on the relative velocity of the colliding pair. In this sense we have to distinguish between:

Maxwell molecules ($\alpha = 0$): The cross section is constant,

$$\sigma_T(r) \equiv \sigma_M := 4\pi C_0 \tag{4.10}$$

This implies that $\bar\sigma(t) \equiv \sigma_M$ at all times. The total rate $R = \frac{N_p\rho\sigma_M}{2\varepsilon}$ is strictly constant and known *a priori*, yielding an exact expected collision count

$$N_c = \frac{N_p\mu\Delta t}{2\varepsilon} \tag{4.11}$$

Where $\mu := \rho\sigma_M$ without inspecting particle velocities.

General VHS models ($\alpha \in (0,1]$): The cross section $\sigma_T(r) = 4\pi C_\alpha r^\alpha$ depends on the relative velocity of the pair. Consequently, the empirical mean of the total cross section $\bar\sigma(t)$ fluctuates as particle velocities evolve, meaning $R(t)$ cannot be evaluated *a priori* without an $O(N_p^2)$ computation over all pairs at every time step, this bottleneck as we will see is resolved by majorization and acceptance-rejection techniques.

4.1.2. Bird's scheme for Maxwell molecules

Because Δt_c in Maxwell molecules is state-independent, the collision process in this case can be simulated by an *exact*, event-driven construction: advance a local "collision clock" t_c in fixed increments Δt_c, and at every tick let a uniformly chosen pair of particles collide. Since the total cross section $\sigma_T \equiv \sigma_M$ is constant, every pair shares the exact same pairwise rate (4.4) and is equally likely to collide next. Consequently, no acceptance-rejection step is required at all, pairs are simply drawn uniformly, e.g. by Algorithm B.6 (sampling without repetition), and the post-collisional velocities are computed from (2.4) with ω sampled uniformly on S^2. The resulting procedure is Algorithm 4.1 below.

Algorithm 4.1 Bird's scheme for Maxwell molecules

---

1. Compute the initial velocities of the particles, $\{v_i^0\}_{i=1,\dots,N_p}$, by sampling them from the initial density $f_0(v)$ (Example B.2.3).
2. Set the collision clock $t_c = 0$ and the (fixed) increment $\Delta t_c = 2\varepsilon/(N_p\mu)$, with $\mu = \rho\sigma_M$.
3. For $n = 0, \dots, n_{\mathrm{TOT}} - 1$:
   - set $\bar{v}_i = v_i^n$ for $i = 1, \dots, N_p$;
   - **repeat**
     - draw a pair (i, j) uniformly among the $N_p(N_p-1)/2$ unordered pairs;
     - sample ω uniformly on S^2 and compute the post-collisional velocities v_i', v_j';
     - set $\bar{v}_i = v_i'$, $\bar{v}_j = v_j'$;
     - update the clock, $t_c \to t_c + \Delta t_c$;

     until $t_c \geq (n+1)\Delta t$;
   - set $v_i^{n+1} = \bar{v}_i$ for $i = 1, \dots, N_p$.

Note that a particle may be selected, and updated, several times within the same macroscopic step $[n\Delta t, (n+1)\Delta t]$, and that each such update uses the *current* working velocity $\bar{v}_i$ rather than the frozen value v_i^n. This is exactly what makes the simulated sequence of pairings a faithful realization of the continuous-time jump process of Section 4.1.1, rather than a time-discretized approximation of it, for *any* choice of Δt, including $\Delta t = t_{\max}$ for a purely space-homogeneous computation.

4.1.3. Bird's scheme for general VHS molecules

As written above, for a general VHS model with $\alpha \in (0, 1]$, σ_T genuinely depends on the relative velocity through (4.9), so $\bar{\sigma}$ in (4.7) is no longer a constant: it is an empirical average over the $N_p(N_p-1)/2$ current pairs, and computing it exactly at every time step would cost $O(N_p^2)$ operations, precisely the cost DSMC is meant to avoid. Bird's idea is to replace the (expensive, exact) uniform sampling of Algorithm 4.1 with an acceptance-rejection step of exactly the kind anticipated, in velocity space, in Section 3.5 and formalized for a discrete index in Algorithm B.5: candidate pairs are still drawn uniformly, but a pair is now accepted as an actual collision with a probability proportional to its own cross section $\sigma_{ij} := \sigma_T(|v_i - v_j|)$, relative to a majorant.

We now evaluate the latter, consider $\bar{v} := N_p^{-1}\sum_i v_i$ and $\Delta v := \max_i |v_i - \bar{v}|$. Then

$|v_i - v_j| \le 2\Delta v$ for every pair i, j, and consequently

$$\Sigma := \sigma_T(2\Delta v) \;\ge\; \sigma_{ij} \qquad \text{for every } 1 \le i < j \le N_p. \tag{4.12}$$

In fact, by the triangle inequality, $|v_i - v_j| = |(v_i - \bar{v}) - (v_j - \bar{v})| \le |v_i - \bar{v}| + |v_j - \bar{v}| \le 2\Delta v$. Since for a VHS potential $\alpha \in [0, 1]$, so $r \mapsto \sigma_T(r) = 4\pi C_\alpha r^\alpha$ is nondecreasing on $[0, \infty)$, and the bound follows.

Now this majorant Σ in (4.12) relies only on Δv, which is computed in just $O(N_p)$ operations per time step. With Σ established, a candidate pair (i, j) drawn uniformly at random is accepted as a physical collision whenever

$$\Sigma\, \xi < \sigma_{ij}, \qquad \xi \sim \mathrm{U}[0, 1]. \tag{4.13}$$

This corresponds precisely to the acceptance-rejection framework of Section 3.5, where the pair is accepted with probability $P_{\mathrm{acc}} = \sigma_{ij}/\Sigma$.

It remains to track the collision clock t_c correctly, now that pairs collide at different rates. Bird's construction assigns to every *accepted* collision of pair (i, j) the local increment

$$\Delta t_{ij} := \frac{2\varepsilon}{N_p\, \rho\, \sigma_{ij}}, \tag{4.14}$$

in place of the (unavailable, ensemble-wide) Δt_c of (4.8). This choice is not ad hoc: since candidates are drawn uniformly and accepted with probability σ_{ij}/Σ, the conditional probability that an accepted collision is precisely the pair (i, j) is

$$p_{ij} = \frac{\sigma_{ij}}{\sum_{1 \le k < l \le N_p} \sigma_{kl}}, \tag{4.15}$$

and a direct computation shows that Δt_{ij} has exactly the right expectation under (4.15):

$$\begin{aligned} \sum_{i<j} \Delta t_{ij}\, p_{ij} &= \sum_{i<j} \frac{2\varepsilon}{N_p \rho \sigma_{ij}} \cdot \frac{\sigma_{ij}}{\sum_{k<l} \sigma_{kl}} = \frac{2\varepsilon}{N_p \rho \sum_{k<l} \sigma_{kl}} \binom{N_p}{2} = \frac{\varepsilon(N_p - 1)}{\rho \sum_{k<l} \sigma_{kl}} \\ &= \frac{\varepsilon(N_p - 1)}{\rho \cdot \frac{N_p(N_p-1)}{2} \bar{\sigma}} = \frac{2\varepsilon}{N_p \rho \bar{\sigma}} = \Delta t_c, \end{aligned} \tag{4.16}$$

using (4.7) in the last line. In other words, even though $\bar{\sigma}$ itself is never computed, advancing the clock by Δt_{ij} upon each accepted collision reproduces, in expectation, the correct mean collision time Δt_c of the current ensemble, and does so *adaptively*, since σ_{ij} in (4.14) is evaluated on the up-to-date velocities of the pair at the moment it is accepted, rather than on a value of $\bar{\sigma}$ frozen at the start of the step. In this way we obtain Bird's algorithm for general VHS molecules, Algorithm 4.2.

Notice that the Maxwell-molecule scheme (Algorithm 4.1) is recovered from Algorithm 4.2 as the degenerate case $\sigma_{ij} \equiv \sigma_M \equiv \Sigma$, where the acceptance test (4.13) succeeds with probability 1 and $\Delta t_{ij} = \Delta t_c$. Furthermore, rather than recomputing Σ at cost $O(N_p)$

Algorithm 4.2 Bird's scheme for general VHS molecules

1. Compute the initial velocities of the particles, $\{v_i^0\}_{i=1,\dots,N_p}$, by sampling them from the initial density $f_0(v)$.
2. Set the collision clock $t_c = 0$.
3. For $n = 0, \dots, n_{\mathrm{TOT}} - 1$:
   - set $\bar{v}_i = v_i^n$ for $i = 1, \dots, N_p$;
   - compute $\Delta v = \max_i |\bar{v}_i - \bar{v}|$ and the majorant $\Sigma = \sigma_T(2\Delta v)$;
   - **repeat**
     - draw a candidate pair (i, j) uniformly among the $N_p(N_p - 1)/2$ unordered pairs (Algorithm B.6);
     - compute $\sigma_{ij} = \sigma_T(|\bar{v}_i - \bar{v}_j|)$ and draw $\xi \sim \mathrm{U}[0, 1]$;
     - **if** $\Sigma\,\xi < \sigma_{ij}$ **then**
       - sample ω uniformly on S^2 and compute the post-collisional velocities v_i', v_j';
       - set $\bar{v}_i = v_i'$, $\bar{v}_j = v_j'$;
       - update the clock, $t_c \to t_c + \Delta t_{ij}$ with Δt_{ij} as in (4.14);

       end if

     until $t_c \geq (n + 1)\Delta t$;
   - set $v_i^{n+1} = \bar{v}_i$ for $i = 1, \dots, N_p$.

every step, one can compute it once at initialization and update it in $O(1)$ time whenever a new pair of post-collisional velocities exceeds the existing bound.

In both cases, for Maxwell molecules and for VHS molecules, the mean collision time Δt_c is proportional to ε. As $\varepsilon \to 0$, the number of collisions $N_c = \Delta t / \Delta t_c$ that Algorithms 4.1 and 4.2 must simulate within a *single* step of the splitting scheme grows like ε^{-1}, so that both schemes become prohibitively expensive precisely in the fluid regime where the Boltzmann equation is best approximated by the much cheaper Euler system. This is the central motivation for the Time Relaxed Monte Carlo schemes.

4.2. Time Relaxed Monte Carlo (TRMC) methods

Time Relaxed Monte Carlo (TRMC) methods, introduced by Pareschi and Russo [3] are constructed to be *asymptotic preserving*, in the precise sense that as $\mu\Delta t/\varepsilon \to \infty$ they degenerate, at no extra cost, into an exact sample from the local Maxwellian equilibrium, rather than requiring more and more simulated collisions. We first lay down the semi-discrete (Wild sum) mathematical background on which every TRMC scheme rests, following [3], and then construct, in increasing order of sophistication, the first-order scheme, its higher-order and the recursive scheme.

4.2.1. The Wild sum and time relaxed schemes

Fixing, as done above, an upper bound Σ of the total cross section σ_T over the velocities under consideration (for Maxwell molecules we may of course take $\Sigma = \sigma_M$, an *exact*

bound), and set

$$\mu := \Sigma\rho. \tag{4.17}$$

We can rewrite the decomposition (4.1), for *any* such μ, as

$$Q(f,f)(v) = P(f,f)(v) - \mu f(v), \tag{4.18}$$

where

$$P(f,f)(v) := Q^+(f,f)(v) + f(v)\int_{\mathbb{R}^3}\big[\Sigma - \sigma_T(|v-v_*|)\big]f(v_*)\,dv_*, \tag{4.19}$$

as one checks by adding and subtracting $\mu f(v) = f(v)\int \Sigma\, f(v_*)\,dv_*$ inside $f(v)\nu[f](v)$. This decomposition is the exact analogue, at the level of the continuum equation, of the acceptance-rejection construction already used for Bird's VHS scheme in Section 4.1.3. Here, P acts as a generalized gain operator (in our framework in which we introduced a majorant to bound the total cross section) encompassing all candidate events, both physical scatterings and *dummy* (fictitious) collisions, so called because rejected candidate pairs undergo zero velocity exchange ($v \to v$). Correspondingly, μf represents the state-independent, Maxwell-like loss rate against which these candidate collisions are evaluated.

In particular, P is a non-negative bilinear operator and for every $f \geq 0$, $f \in L^1(\mathbb{R}^3)$ and every collision invariant $\phi(v) = 1, v, |v|^2$,

$$\int_{\mathbb{R}^3} P(f,f)(v)\,\phi(v)\,dv = \mu\int_{\mathbb{R}^3} f(v)\,\phi(v)\,dv. \tag{4.20}$$

The non-negativity of P is immediate: $Q^+(f,f) \geq 0$ as an integral of non-negative quantities, and the bracketed difference in the correction term of (4.19) satisfies $\Sigma - \sigma_T(|v-v_*|) \geq 0$ everywhere on the support of f by construction of the majorant Σ.

To verify (4.20), we rearrange (4.18) to express $P(f,f) = Q(f,f) + \mu f(v)$. Multiplying by a collision invariant $\phi(v)$ and integrating over $\mathbb{R}^3$ yields

$$\int_{\mathbb{R}^3} P(f,f)(v)\,\phi(v)\,dv = \int_{\mathbb{R}^3} Q(f,f)(v)\,\phi(v)\,dv + \mu\int_{\mathbb{R}^3} f(v)\,\phi(v)\,dv.$$

Recalling the fundamental conservation property of the collision operator, $\int_{\mathbb{R}^3} Q(f,f)\phi\,dv = 0$, the first integral vanishes on the right-hand side, leaving exactly (4.20).

The Wild sum

With P and μ as above, the space-homogeneous problem (2.16) becomes

$$\frac{\partial \bar{f}}{\partial t} = \frac{1}{\varepsilon}\big[P(\bar{f},\bar{f}) - \mu\bar{f}\big]. \tag{4.21}$$

Following Wild [6], it is possible to introduce the time change

$$\tau = 1 - e^{-\mu t/\varepsilon} \in [0,1), \qquad F(v,\tau) := \bar{f}(v,t)\,e^{\mu t/\varepsilon}. \tag{4.22}$$

A direct computation, using the bilinearity (hence quadratic homogeneity) of P and $d\tau/dt = (\mu/\varepsilon)e^{-\mu t/\varepsilon}$, gives

$$\frac{\partial F}{\partial \tau} = \frac{1}{\mu} P(F, F), \qquad F(v, 0) = f_0(v). \tag{4.23}$$

Equation (4.23) can be solved by the ansatz of a power series in τ,

$$F(v, \tau) = \sum_{k=0}^{\infty} \tau^k f_k(v), \qquad f_0(v) = f_0(v), \tag{4.24}$$

(the two occurrences of f_0 deliberately coincide: the $k = 0$ coefficient of the series is exactly the initial datum). Substituting (4.24) into (4.23), matching powers of τ on both sides it is possible to obtain the explicit recurrence relation:

$$f_{k+1}(v) = \frac{1}{k+1} \sum_{h=0}^{k} \frac{1}{\mu} P(f_h, f_{k-h})(v), \qquad k = 0, 1, \ldots, \tag{4.25}$$

which determines every f_k explicitly, and recursively, from f_0 alone and using only basic evaluations of the bilinear gain operator P. Undoing the change of variables (4.22) in equation 4.24 yields the formal Wild sum representation of the solution for the collision step 2.16,

$$\bar{f}(v, t) = e^{-\mu t/\varepsilon} \sum_{k=0}^{\infty} \left(1 - e^{-\mu t/\varepsilon}\right)^k f_k(v). \tag{4.26}$$

Wild [6] originally introduced this series representation exclusively for the Boltzmann equation with Maxwell molecules, where the total cross section is constant and the collision frequency $\nu[f](v) \equiv \mu$ is naturally state-independent. The derivation presented above, following Pareschi and Russo [3], demonstrates that the expansion extends to general VHS models once fictitious (dummy) collisions are introduced to establish a state-independent majorant loss rate $\mu = \rho\Sigma$.

In fact, while a velocity-dependent frequency $\mu(v)$ would formally allow a similar power series derivation, $\mu(v)$ could no longer be factored out of the integral in the conservation property $\int_{\mathbb{R}^3} P(f, f)\phi \, dv = \int_{\mathbb{R}^3} \mu(v) f \phi \, dv$. Consequently, truncating the Wild sum in practical numerical schemes would break the exact discrete conservation of mass, momentum, and energy. Enforcing a strictly constant μ via (4.17) guarantees that the transition operator $\mu^{-1}P(f, f)$ preserves collision invariants exactly at every truncation order.

Truncation and the class of TR schemes

Truncating (4.26) at a finite order $m \geq 1$ and replacing the (unknown, infinite) tail by the stationary solution of (2.16), the local Maxwellian $M = M(\rho, u, T)$ associated with the current moments of $\bar{f}$, gives the class of *Time Relaxed* (TR) schemes: with $f^n(v) \approx \bar{f}(v, n\Delta t)$,

$$f^{n+1}(v) = e^{-\mu \Delta t/\varepsilon} \sum_{k=0}^{m} \left(1 - e^{-\mu \Delta t/\varepsilon}\right)^k f_k^n(v) + \left(1 - e^{-\mu \Delta t/\varepsilon}\right)^{m+1} M(v), \tag{4.27}$$

where f_k^n is obtained from $f_0^n = f^n$ through (4.25) applied at time $n\Delta t$.

We now examine the properties of scheme (4.27).

i) *Conservation:* If $f^n \geq 0$, then $f^{n+1} \geq 0$ for every $\mu\Delta t/\varepsilon > 0$, and

$$\int_{\mathbb{R}^3} f^{n+1}(v)\,\phi(v)\,dv = \int_{\mathbb{R}^3} f^n(v)\,\phi(v)\,dv, \qquad \phi(v) = 1, v, |v|^2.$$

ii) *Asymptotic preservation, AP:* For every $m \geq 1$, $\lim_{\mu\Delta t/\varepsilon \to \infty} f^{n+1} = M(v)$.

To check the validity of these two properties we now first establish that the update is a valid convex combination of moment-preserving, non-negative distributions, and then we will look at the limit $\mu\Delta t/\varepsilon \to \infty$.

Step 1: Positivity and Moment Conservation. We first verify by induction that each recursively defined term f_k^n in the Wild sum is non-negative and preserves the physical moments of f^n.

Base case $k = 0$: By definition, $f_0^n = f^n \geq 0$, so non-negativity and moment equality hold trivially. Inductive step: Assume $f_h^n \geq 0$ and $\int_{\mathbb{R}^3} f_h^n \phi\,dv = \int_{\mathbb{R}^3} f^n \phi\,dv$ for all $0 \leq h \leq k$, where $\phi(v) \in \{1, v, |v|^2\}$. Since P is a non-negative operator, f_{k+1}^n is defined in (4.25) as a sum of products of non-negative functions, ensuring $f_{k+1}^n \geq 0$. Integrating f_{k+1}^n against $\phi(v)$ and applying the bilinear conservation property of P, namely $\int_{\mathbb{R}^3} P(f,g)\phi\,dv = \mu \int_{\mathbb{R}^3} f\phi\,dv$, yields:

$$\int_{\mathbb{R}^3} f_{k+1}^n(v)\,\phi(v)\,dv = \frac{1}{k+1}\sum_{h=0}^{k} \frac{1}{\mu}\int_{\mathbb{R}^3} P(f_h^n, f_{k-h}^n)(v)\,\phi(v)\,dv = \frac{1}{k+1}\sum_{h=0}^{k}\int_{\mathbb{R}^3} f_h^n(v)\,\phi(v)\,dv.$$

By the inductive hypothesis, each integral inside the sum equals $\int_{\mathbb{R}^3} f^n\phi\,dv$. Averaging over the $k+1$ identical terms recovers:

$$\int_{\mathbb{R}^3} f_{k+1}^n(v)\,\phi(v)\,dv = \int_{\mathbb{R}^3} f^n(v)\,\phi(v)\,dv.$$

Furthermore, the local Maxwellian $M(v)$ is non-negative by definition and constructed using the exact macroscopic moments of f^n, so $\int_{\mathbb{R}^3} M\phi\,dv = \int_{\mathbb{R}^3} f^n\phi\,dv$.

Next, setting $x = e^{-\mu\Delta t/\varepsilon} \in (0,1]$, the weights in scheme (4.27) are $w_k = x(1-x)^k$ for $k = 0, \ldots, m$ and $w_M = (1-x)^{m+1}$ for the Maxwellian tail. Using the geometric identity $\sum_{k=0}^{m} x(1-x)^k = 1 - (1-x)^{m+1}$, the weights sum to unity:

$$\sum_{k=0}^{m} w_k + w_M = 1 - (1-x)^{m+1} + (1-x)^{m+1} = 1.$$

Because $x \in (0,1]$, every weight w_k and w_M is strictly positive. Thus, f^{n+1} is expressed as a convex combination of non-negative functions $\{f_0^n, \ldots, f_m^n, M\}$ that each share the exact same first three moments with f^n. It follows directly that $f^{n+1} \geq 0$ and $\int_{\mathbb{R}^3} f^{n+1}\phi\,dv = \int_{\mathbb{R}^3} f^n\phi\,dv$.

Step 2: Asymptotic Preservation (AP). In the hydrodynamic regime ($\varepsilon \to 0$ or $\mu\Delta t/\varepsilon \to \infty$), $x = e^{-\mu\Delta t/\varepsilon} \to 0$. As a result, each of the $m+1$ head weights vanishes, $w_k = x(1-x)^k \to 0$, whereas the Maxwellian tail weight approaches unity, $w_M = (1-x)^{m+1} \to 1$ completing the proof.

Generalized weights

The construction above is one instance of a more flexible family of truncated sum obtained by the Wild sum. In general we can write:

$$f^{n+1} = \sum_{k=0}^{m} A_k f_k^n + A_{m+1} M, \tag{4.28}$$

with weights $A_k = A_k(\tau)$, $\tau = 1 - e^{-\mu\Delta t/\varepsilon}$, the 2 steps proof just evaluated remains valid also in this case under the following three conditions on the A_k.

Let $A_k = A_k(\tau) \geq 0$, $k = 0, \dots, m+1$, satisfy

$$\textit{(consistency)} \quad \lim_{\tau\to 0} A_1(\tau)/\tau = 1, \qquad \lim_{\tau\to 0} A_k(\tau)/\tau = 0, \quad k = 2, \dots, m+1, \tag{4.29}$$

$$\textit{(conservation)} \quad \sum_{k=0}^{m+1} A_k(\tau) = 1, \qquad \tau \in [0,1], \tag{4.30}$$

$$\textit{(AP)} \quad \lim_{\tau\to 1} A_k(\tau) = 0, \qquad k = 0, \dots, m. \tag{4.31}$$

Notice that the choice

$$A_k = (1-\tau)\tau^k, \quad k = 0, \dots, m, \qquad A_{m+1} = \tau^{m+1}, \tag{4.32}$$

reproduces the truncated Wild sum scheme (4.27). A different example instead, which will be convenient for the recursive construction of Section 4.2.4,

$$A_k = (1-\tau)\tau^k, \quad k = 0, \dots, m-1, \qquad A_m = 1 - \sum_{k=0}^{m-1} A_k - A_{m+1}, \qquad A_{m+1} = \tau^{m+2}, \tag{4.33}$$

corresponds to setting $f_{m+1} = f_m$ and $f_k = M$ for $k \geq m+2$ in (4.26). However determining the set of weights that minimizes the statistical error of the resulting scheme, for fixed m, remains an open problem, as [3] notice.

4.2.2. First-order TRMC for VHS molecules

Taking $m = 1$ in (4.27), or equivalently in (4.32), gives the first-order scheme

$$f^{n+1} = A_0 f^n + A_1 f_1 + A_2 M, \qquad A_0 = 1-\tau, \quad A_1 = (1-\tau)\tau, \quad A_2 = \tau^2, \tag{4.34}$$

Since, as we have seen, $A_0 + A_1 + A_2 = 1$ the equation (4.34) admits a direct probabilistic reading: a particle sampled from f^n does not collide with probability A_0; undergoes a

single candidate collision step with another particle sampled from f^n with probability A_1 (updating its state according to $f_1 = \mu^{-1}P(f^n, f^n)$ via standard majorized acceptance–rejection); or it is discarded and replaced by a fresh sample from the Maxwellian with probability A_2.

If we now approximate scheme (4.34) using a finite ensemble of N_p simulation particles, sampling individual particles independently guarantees conservation of mass, momentum, and energy only in expectation. For finite N_p, independent single-particle choices introduce stochastic noise that causes artificial numerical drift over time.

To enforce exact, pointwise conservation across the ensemble at every time step, two modifications are made to the discrete particle mechanics. First, candidate collisions are performed on uniformly selected particle *pairs* rather than individuals, ensuring microscopic binary momentum and energy conservation during collision updates. Second, the sub-ensemble selected for Maxwellian relaxation is replaced by drawing fresh velocity vectors using Pullin's algorithm [4]. This method conditions the generated Maxwellian on the *exact empirical total momentum and kinetic energy* of the specific subset of particles being replaced, preventing statistical thermal drift.

These choices lead to the fully conservative first-order TRMC particle algorithm.

Algorithm 4.3 First-order TRMC for VHS molecules

---

1. Compute the initial velocities of the particles, $\{v_i^0\}_{i=1,\dots,N_p}$, by sampling them from the initial density $f_0(v)$.
2. For $n = 1, \dots, n_{\mathrm{TOT}}$:
   - Compute a state-independent majorant cross section Σ and set $\mu = \rho\Sigma$ as done in Bird's scheme.
   - Evaluate $\tau = 1 - \exp(-\mu\Delta t/\varepsilon)$ and compute weights $A_1(\tau) = (1-\tau)\tau$ and $A_2(\tau) = \tau^2$.
   - Set $N_c = \mathrm{Iround}(N_p A_1/2)$. Uniformly select $2N_c$ distinct particles without replacement to form N_c pairs, and execute candidate collision steps using majorized acceptance–rejection (as in Algorithm 4.2).
   - Maxwellian Thermalization: Set $N_M = \mathrm{Iround}(N_p A_2)$. From the remaining $N_p - 2N_c$ uncollided particles, select N_M particles without replacement. Compute their empirical total momentum P_{sub} and total kinetic energy E_{sub}.
   - Replace these N_M selected particles with fresh velocity vectors generated via Pullin's matching algorithm [4], preserving P_{sub} and E_{sub} exactly.
   - Set $v_i^{n+1} = v_i^n$ for the remaining $N_0 = N_p - 2N_c - N_M$ untouched particles.

4.2.3. Higher-order TRMC schemes

Truncating (4.26) one order later, at $m = 2$ gives the second-order scheme

$$f^{n+1} = A_0 f^n + A_1 f_1 + A_2 f_2 + A_3 M, \qquad f_1 = \frac{P(f^n, f^n)}{\mu}, \quad f_2 = \frac{P(f^n, f_1)}{\mu}, \tag{4.35}$$

with $A_0, \dots, A_3$ given by (4.32) at $m = 2$. Its probabilistic reading extends that of the first-order scheme term by term: a fraction A_0 of the particles does not collide; a fraction

A_1 is sampled from f_1 exactly as before; a new fraction A_2 instead is sampled from f_2, i.e. from a *second-generation* collision in which particles freshly sampled from f^n, is made to collide with particles that were themselves the outcome of a first-generation collision, and a fraction A_3 is sampled again from the Maxwellian.

A third-order scheme is obtained in exactly the same way, keeping one more term of the Wild sum,

$$f^{n+1} = A_0 f^n + A_1 f_1 + A_2 f_2 + A_3 f_3 + A_4 M, \qquad f_3 = \frac{1}{3\mu}\big[2P(f^n, f_2) + P(f_1, f_1)\big], \tag{4.36}$$

by (4.25) at $k = 2$; the resulting particle algorithm follows the same pattern of nested collisions between successive generations and is not detailed further here. In practice, direct evaluation of higher-order terms is computationally prohibitive due to the exponential growth of collision histories. The following section analyzes these bottlenecks in detail and introduces an efficient framework to compute higher-order generations.

4.2.4. Recursive TRMC methods

The Wild sum as a collision genealogy: Maxwell molecules

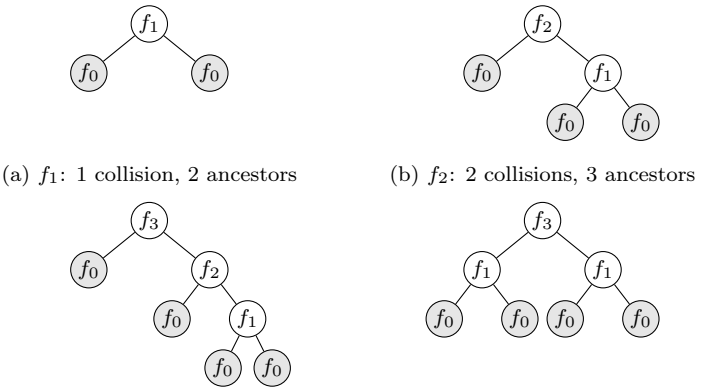
To understand the computational disadvantages associated with evaluating higher-order generations, consider the spatially homogeneous Boltzmann equation for Maxwell molecules. In this regime, the exact solution over a single time step Δt can be represented via the Wild sum expansion (4.26), which admits a direct probabilistic interpretation.

When picking a particle at random from the velocity distribution $\bar{f}(\cdot, \Delta t)$, the probability that it has undergone zero collisions during $[0, \Delta t]$ is given by the weight of f_0, namely $e^{-\mu \Delta t/\varepsilon}$. Conditioned on this event, its velocity is distributed according to the initial density f_0. Similarly, the probability that the particle has experienced *exactly one* collision is given by the coefficient $e^{-\mu \Delta t/\varepsilon}(1 - e^{-\mu \Delta t/\varepsilon})$. Under this condition, its velocity is distributed according to $f_1 = \mu^{-1} P(f_0, f_0)$, which corresponds to the outcome of a single binary collision between two independent ancestors drawn from f_0.

More generally, the term f_n in the Wild recurrence (4.25) represents the conditional velocity distribution of a particle whose backward collision history back to $t = 0$ forms a binary tree containing exactly n internal collision events and $n + 1$ original uncollided ancestor particles from f_0.

As illustrated in Figure 4.1, generating a sample from f_n recursively requires evaluating the collision of two sub-trees: one of order k and one of order $n - k - 1$. Since every binary collision combines two incoming branches into one outgoing state, a tree with n internal collision nodes strictly requires $n + 1$ leaf samples from f_0. Without storing or caching intermediate generations f_k, evaluating f_n forces a complete traversal down to all $n + 1$ leaves.

Leveraging this probabilistic interpretation allows us to formulate a direct recursive sampling algorithm for $\bar{f}(\cdot, \Delta t)$ without advancing the global particle system through sequen-



(a) f_1: 1 collision, 2 ancestors

(b) f_2: 2 collisions, 3 ancestors

(c) f_3 (asymmetric $P(f_0, f_2)$): 3 collisions, 4 ancestors

(d) f_3 (symmetric $P(f_1, f_1)$): 3 collisions, 4 ancestors

Figure 4.1: Binary collision trees (genealogies) for generations f_1– f_3. Internal white nodes denote binary collision events $P(\cdot, \cdot)$, and shaded leaf nodes denote initial uncollided particles drawn from f_0. Every generation f_n requires n internal collision nodes and exactly $n+1$ original leaf particles, regardless of whether the tree topology is asymmetric (as in c) or symmetric (as in d).

tial time steps.

Algorithm 4.4 Recursive sampling for Maxwell molecules

1. Choose n from a geometric distribution with parameter $\tau = 1 - e^{-\mu \Delta t/\varepsilon}$, note $\mathbb{E}[n] = e^{\mu \Delta t/\varepsilon} - 1$ which grows rapidly with $\mu \Delta t/\varepsilon$.
2. Take a sample from the density f_n:
   - if $n = 0$, take v from the initial density f_0;
   - otherwise, choose $k \in \{0, 1, \ldots, n-1\}$ with equal probability, take a sample v_i from the density f_k and a sample v_j from the density f_{n-k-1} both, recursively, as in this step 2, perform the collision between v_i and v_j, and return $v = v_i'$, which is distributed exactly according to f_n.

If we iterate this algorithm for N_p times, once at a time, we obtain the desired ensemble of simulation particles used to approximate $\bar{f}(\cdot, \Delta t)$, and it is straightforward to implement. However, it should be noted that the recursion in Step 2 terminates in at most n nested levels, which exactly correspond to the depth in the fully asymmetric tree topology where each level of the tree corresponds to a collision. Thus, as some might expect, direct reliance on Algorithm 4.4 presents major practical drawbacks. The expected tree size $\mathbb{E}[n] = e^{\mu \Delta t/\varepsilon} - 1$ grows exponentially in both Δt and ε^{-1}. Consequently, both computation time and the memory requirements explode near the fluid regimes ($\varepsilon \to 0$). Moreover, because each particle's collision tree is sampled independently of the others, no shared history exists across the ensemble, leading to stochastic fluctuations that violate exact momentum and energy conservation.

Recursive TRMC for VHS molecules

To eliminate these drawbacks, we generate a mutually consistent population of N_p particles simultaneously at time Δt. By exploiting the known convergence of the Wild sum expansion toward the Maxwellian distribution, we cap the recursion depth at a finite integer m. Truncating the expansion according to (4.27), the target population is partitioned into:

- $N_0 = \mathrm{Iround}\big(N_p(1-\tau)\big)$ particles sampled from f_0,
- $N_n = \mathrm{Iround}\big(N_p(1-\tau)\tau^n\big)$ particles sampled from f_n, for $1 \le n \le m$,
- $N_{m+1} = N_p - \sum_{n=0}^{m} N_n$ particles sampled from the equilibrium Maxwellian,

where $\tau = 1 - e^{-\mu\Delta t/\varepsilon}$. Because $(1-\tau)\tau^n$ is a strictly decreasing function of n, particle allocation across generations is non-increasing. Consequently, the truncation depth m is capped automatically as soon as the running sum $\bar{N} = \sum_{k=0}^{n} N_k$ exceeds N_p or the required increment N_n drops below a single particle.

As we noticed, sampling each generation f_n independently via isolated tree evaluations is computationally wasteful and violates exact conservation laws, as identical ancestor f_k-particles ($k < n$) are regenerated repeatedly. To resolve this, the entire population is constructed as a single, shared collision tree.

Generating an initial pair of f_m-particles requires $m+1$ particles sampled from the initial distribution f_0. These $m+1$ particles are preserved throughout the collision sequence to guarantee exact conservation of physical quantities (mass, momentum, and energy). Figure 4.2 illustrates this process for $m = 5$, where generating the top f_5 pair from $m+1 = 6$ leaf particles from f_0 yields intermediate post-collision particles distributed according to f_2 and f_1.

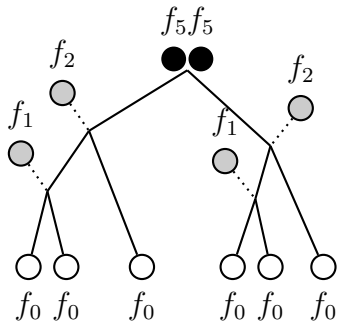


Figure 4.2: Generation of particles via binary collision tree in the TRMC algorithm.

For each binary collision, one resulting particle ascends the tree to fulfill the parent collision, while its twin companion is stored in a buffer. Storage counters c_n track the number of available buffered particles for each generation (c_1 for f_1, c_2 for f_2, etc.).

Generating the target N_m particles for f_m populates these intermediate buffers, but does not guarantee that lower generations N_n ($n < m$) are fully satisfied. Thus, the algorithm proceeds iteratively from $n = m$ down to $n = 1$. Whenever a collision partner from

generation k is required, the algorithm checks whether $c_k > 0$. If stored particles exist, one is selected at random, c_k is decremented, and the target count N_k is adjusted. If no stored particle is available ($c_k = 0$), the particle is generated recursively. Finally, to handle the remaining N_{m+1} tail particles while guaranteeing exact conservation of mass, momentum, and energy, N_{m+1} particles are initially drawn from f_0 to measure their sample fluid moments. Their initial non-equilibrium velocities are then discarded and replaced by N_{m+1} new particles sampled from an equilibrium Maxwellian matching those exact computed moments (e.g., using Pullin's algorithm), analogously to the first-order TRMC method in Algorithm 4.3.

Algorithm 4.5 Recursive TRMC for VHS molecules

1. Compute the initial velocities $\{v_i^0\}_{i=1,\dots,N_p}$ by sampling them from $f_0(v)$.
2. Set $\tau = 1 - \exp(-\rho\Sigma\Delta t/\varepsilon)$ (with Σ an upper bound of the cross section as before), $N_0 = \mathrm{Iround}\big(N_p(1-\tau)\big)$, $\bar{N} = N_0$, $n = 0$.
3. **repeat**: set $n \to n+1$; compute $N_n = \mathrm{Iround}\big(N_p(1-\tau)\tau^n\big)$; set $\bar{N} \to \bar{N} + N_n$; **until** $\bar{N} > N_p$ or $N_n < 1$.
4. Set $m = n - 1$ and $N_{m+1} = N_p - (\bar{N} - N_n)$ (the particles not yet accounted for by generations $0, \dots, m$).
5. Set counters $c_n = 0$ for $n = 1, \dots, m+1$, and compute an upper bound Σ of σ_{ij} as in DSMC.
6. For $n = m, \dots, 1$: draw N_n samples from f_n, repeating the following until N_n new particles have been produced:
   - choose $k \in \{0, \dots, n-1\}$ with equal probability;
   - if $k = 0$, take v_i from f_0; otherwise, if $c_k > 0$, reuse a stored f_k-particle at random (decrement c_k, increment the remaining target N_k); otherwise sample v_i and a companion v_i^* from f_k recursively, store v_i^* and increment c_k (decrement N_k);
   - symmetrically, if $n - k - 1 = 0$ take v_j from f_0, else obtain v_j from f_{n-k-1} by the same stored/recursive rule;
   - if $\Sigma\xi < \sigma_{ij}$, perform the collision between v_i and v_j as in DSMC; the outcome v_i', v_j' is distributed according to f_n; decrement N_n by 2.
7. Sample the remaining N_{m+1} particles from the Maxwellian, exactly as we did in Algorithm 4.3.

In practice, m can be rather large; it is customary to fix a maximum allowed tree depth $m_{\max}$ at the start of the computation. Choosing m smaller than prescribed by (4.2.4) yields a faster but less accurate algorithm, equivalent to truncating higher-order direct schemes at a lower order. For space-inhomogeneous problems where uniform accuracy in time is critical, the recursive scheme, where tree depth can be adapted cell-by-cell to the local value of $\mu\Delta t/\varepsilon$, represents the most promising formulation presented in this chapter, as evaluated by [3].

4.3. Generalization to the space inhomogeneous case

We close the chapter by returning, as promised, to the full space-inhomogeneous Boltzmann equation (2.1), and by showing how any of the space-homogeneous particle schemes constructed above, namely Bird's scheme or any of the TRMC variants, supplies the collision half-step of the operator splitting (2.16)–(2.17) of Section 2.4.

4.3.1. The splitting particle method

The idea is particularly simple. The physical domain Ω is partitioned into N_{cell} cells; writing N_J for the number of simulation particles currently located in cell J, the collision step over $[n\Delta t, (n+1)\Delta t]$ is solved, independently in every cell, by Monte Carlo, under the approximation that the density is spatially constant within each cell, precisely the space-homogeneous approximation underlying (2.16), so that a particle may collide only with another particle currently located in the same cell. Any of the schemes of Sections 4.1–4.2 may be used for this local update, with the cell's own particle count N_J in the role of N_p and its local number density

$$\rho_J = N_J\, m^* \tag{4.37}$$

in the role of ρ, where m^* is the (fixed) mass represented by a single simulation particle, in general much larger than the mass of a single physical molecule, since a simulation reproduces a physical system of very many molecules with a much smaller number of computational particles.

In the spatially inhomogeneous case, prior to performing the local collision step, each particle must be located and mapped to its enclosing spatial cell. This spatial indexing task can be accomplished in linear time, $O(N_p)$, using bin-sorting or linked-list data structures. Specifically, one constructs two index arrays, `kp` and `cp`, such that the particles contained in cell J are precisely those indexed by $\texttt{kp}[k]$ for

$$\texttt{cp}[J-1] < k \le \texttt{cp}[J].$$

Following the local collision phase, given the particle positions x_i^n at time $t_n = n\Delta t$ and the post-collision velocities $\bar{v}_i$ ($i = 1, \dots, N_p$), the convection step is executed via a single global loop over all particles:

$$\bar{x}_i = x_i^n + \bar{v}_i \Delta t, \qquad i = 1, \dots, N_p. \tag{4.38}$$

In the absence of physical boundaries, the updated states at t_{n+1} are assigned directly as $x_i^{n+1} = \bar{x}_i$ and $v_i^{n+1} = \bar{v}_i$.

In matters of the choice of the time-step Δt event-driven approaches such as Bird's algorithm utilize a collision clock which, as we have seen, have a computational cost that grows rapidly and becomes prohibitive for fixed Δt in near-continuum regimes ($\varepsilon \to 0$).

Conversely, TRMC schemes preserve a strictly bounded cost of $O(N_p)$ operations per cell for any value of $\mu\Delta t/\varepsilon$, including the fluid limit $\mu\Delta t/\varepsilon \to \infty$. This asymptotic robustness makes TRMC especially powerful for space-inhomogeneous problems characterized by widely varying local Knudsen numbers.

4.3.2. Implementation of boundary conditions

We now show how the boundary conditions of Section 2.2 are realized at the particle level; we restrict, for the sake of simplicity, to a one-dimensional domain $\Omega = [0, L]$, the extension to several dimensions being straightforward. All the boundary conditions that follows are applied during the convection step, after the candidate positions $\bar{x}_i$ of (4.38) have been computed.

Periodic boundaries

In the simplest case, a candidate position that has left $[0, L]$ is translated back into the domain by one period.

Algorithm 4.6 Periodic boundary conditions

For $i = 1, \dots, N_p$: if $0 < \bar{x}_i < L$, set $x_i^{n+1} = \bar{x}_i$, $v_i^{n+1} = \bar{v}_i$; if $\bar{x}_i < 0$, set $x_i^{n+1} = \bar{x}_i + L$, $v_i^{n+1} = \bar{v}_i$; if $\bar{x}_i > L$, set $x_i^{n+1} = \bar{x}_i - L$, $v_i^{n+1} = \bar{v}_i$.

Reflecting walls

Under pure specular reflection, which correspond to the $\alpha = 0$ case of the Maxwell boundary condition 2.8, any particle exiting the domain is handled simply by mirroring its position back across the boundary and inverting the normal component of its velocity.

Algorithm 4.7 Reflecting boundary conditions

For $i = 1, \dots, N_p$: if $0 < \bar{x}_i < L$, set $x_i^{n+1} = \bar{x}_i$, $v_i^{n+1} = \bar{v}_i$; if $\bar{x}_i < 0$ or $\bar{x}_i > L$, set $x_i^{n+1} = b - |\bar{x}_i - b| \operatorname{sgn}(\bar{v}_{i,x})$, $v_{i,x}^{n+1} = -\bar{v}_{i,x}$, $v_{i,y}^{n+1} = \bar{v}_{i,y}$, $v_{i,z}^{n+1} = \bar{v}_{i,z}$, where b is 0 or L correspond to the boundary crossed.

Maxwell (thermal accommodation) boundaries

The general boundary condition 2.8, with accommodation coefficient $\alpha \in (0, 1)$, mixes specular reflection with diffuse thermal re-emission: with probability $1 - \alpha$ the particle undergoes the specular reflection of Algorithm 4.7; with probability α it is discarded and re-emitted from the wall with a velocity sampled from the half-Maxwellian $M_w(v)$, $v \cdot n > 0$, of (2.9), corresponding to a gas in thermodynamic equilibrium with the wall at temperature T_w.

Algorithm 4.8 Maxwell boundary conditions, at $\bar{x}_i < 0$

1. Draw $\xi \sim \mathrm{U}[0, 1]$; if $\xi > \alpha$), apply Algorithm 4.7 and stop.
2. Otherwise, compute $\Delta\tilde{t} = |\bar{x}_i / \bar{v}_{i,x}|$, the time at which the (straight-line) trajectory would have crossed the wall.
3. Sample a velocity $\tilde{v}$ from the half-Maxwellian $M_w(v)$, $v_x > 0$ (e.g. by composition and inversion, Appendix B.1.3, on the flux-weighted density $v_x M_w(v)$), and set $v_{i,x}^{n+1} = |\tilde{v}_x|$, $v_{i,y}^{n+1} = \tilde{v}_y$, $v_{i,z}^{n+1} = \tilde{v}_z$.
4. Set $x_i^{n+1} = v_{i,x}^{n+1} \, \Delta\tilde{t}$.

An analogous procedure applies at the wall $x = L$.

Inflow boundaries

Finally, the inflow condition 2.10, $f(x, v, t) = g(v, t)$ on the incoming part of the boundary, is realized by injecting, at every time step, new particles at the domain's edges, sampled from the flux-weighted density $|v_x|\, g(v, t)$; if g is itself a Maxwellian $M(\rho_L, u_L, T_L)$ at the left boundary, the expected number of particles entering during Δt through a boundary patch of area S is

$$\langle N_L^{\text{new}} \rangle = S\, \Delta t \int_{v_x>0} v_x\, M(\rho_L, u_L, T_L)(v)\, dv, \tag{4.39}$$

and each injected particle, since it is assumed that they are generated at a time uniformly distributed within $[n\Delta t, (n+1)\Delta t]$, is placed at $x^* = b + v_x^* \Delta t\, \xi$, $\xi \sim \mathrm{U}[0, 1]$, where $b \in \{0, L\}$, again, is the boundary through which it entered. Furthermore, if exact particle conservation within the computational domain is required deterministically rather than in expectation, one can record the precise number of exiting particles during a time step and re-inject an identical count through the inflow boundary.

This completes the particle framework constructed throughout this chapter and closes the circle opened in Chapter 2 with the description and derivation of the Boltzmann equation itself.

I hope this report gave a clear, self-contained account of Pareschi and Russo's framework, together with the derivations and extensions I found useful to add along the way. If you have any questions, constructive feedback, or ideas for further discussion, feel free to reach out at fogarintommaso@gmail.com

Bibliography

[1] N. H. Johannesen. Molecular gas dynamics. by g. a. bird. oxford university press, 1976.289 pp. £212.00. *Journal of Fluid Mechanics*, 80(3):607–607, 1977. doi: 10.1017/S0022112077221860.

[2] G. E. Lee. The boltzmann equation. `https://gyu-eun-lee.github.io/academic/gso_boltzmann.pdf`, 2020. Notes for a GSO talk, December 2019; last updated January 4, 2020.

[3] L. Pareschi and G. Russo. An introduction to monte carlo method for the boltzmann equation. *ESAIM: Proceedings*, 10:35–75, 08 2002. doi: 10.1051/proc:2001004.

[4] D. I. Pullin. Generation of normal variates with given sample mean and variance. 9 (4):303–309, 1979. doi: 10.1080/00949657908810333.

[5] S. Weinzierl. Introduction to monte carlo methods. *ArXiv High Energy Physics - Phenomenology e-prints*, 07 2000. doi: 10.1007/978-0-387-87837-9_1.

[6] E. Wild. On boltzmann's equation in the kinetic theory of gases. *Mathematical Proceedings of the Cambridge Philosophical Society*, 47:602 – 609, 1951. URL `https://api.semanticscholar.org/CorpusID:124530011`.

A | Appendix A

Following the approach of Lee [2] and the lectures of Prof. Silvia Lorenzani, this appendix gives an intuitive derivation of the Boltzmann equation from the underlying N-particle dynamics, closely mirroring Boltzmann's original physical reasoning. Rather than aiming for the mathematically rigorous justification our goal is to make explicit the physical hypotheses on which the equation rests. As we point out along the way, these are precisely the hypotheses that justify the particle-based Monte Carlo methods discussed in the main text.

A.1. Problem Setting

Consider a system of N identical hard-sphere particles of radius r in $\mathbb{R}^3$, subject to no external forces. Two particles collide exactly when the distance between their centers equals the contact distance

$$\sigma := 2r. \tag{A.1}$$

Between collisions each particle moves in a straight line at constant velocity (Newton's first law), and all collisions are assumed to be perfectly elastic, conserving total momentum and total kinetic energy.

We nevertheless assume that our gas of hard-spheres satisfies the:

(A1) **Rarefied gas regime.** The number of particles N is large, but the gas is dilute enough that

$$N\sigma^3 \ll 1, \qquad N\sigma^2 \sim 1. \tag{A.2}$$

The first condition ensures that the volume occupied by the particles is negligible compared to the volume of the domain, so that three-body and higher-order encounters can be neglected in favor of binary collisions; the second fixes the mean free path to be of order one, comparable to the macroscopic length scale, so that particles do undergo a nontrivial number of collisions.

Because the full $6N$-dimensional Hamiltonian system is intractable for $N \gg 1$, we abandon the idea of tracking every particle individually and instead describe the gas statistically, through the one-particle marginal distribution $P^{(1)}(t, x, v)$: the probability density of finding a single, tagged particle at position x with velocity v at time t. Our goal in what follows is to derive a closed evolution equation for $P^{(1)}$ as $N \to \infty$.

A.2. Free Transport and the Gain–Loss Structure

In the absence of collisions, particles undergo free transport along straight trajectories, and $P^{(1)}$ satisfies the free transport equation

$$\partial_t P^{(1)} + v \cdot \nabla_x P^{(1)} = 0. \tag{A.3}$$

Collisions modify this balance by adding a gain term G and a loss term L,

$$\partial_t P^{(1)} + v \cdot \nabla_x P^{(1)} = G - L, \tag{A.4}$$

where, loosely speaking, $L(t,x,v)\,\mathrm{d}x\,\mathrm{d}v\,\mathrm{d}t$ counts the particles removed from the infinitesimal phase-space cell $[x, x+\mathrm{d}x] \times [v, v+\mathrm{d}v]$ by a collision occurring in $[t, t+\mathrm{d}t]$, while $G(t,x,v)\,\mathrm{d}x\,\mathrm{d}v\,\mathrm{d}t$ counts the particles entering that same cell through a collision in the same time interval. Determining G and L requires analyzing the mechanics of a single binary collision, which we now do.

A.3. Kinematics of an Elastic Binary Collision between Hard Spheres

Consider two particles with pre-collisional phase-space coordinates (x,v) and (x_*,v_*), and post-collisional coordinates (x',v') and (x'_*,v'_*). Two further hypotheses enter the description of the collision itself.

(A2) **Locality and instantaneity.** A collision takes place at a single point in space and has negligible duration, so that the two particles do not influence each other's trajectories except at the instant of contact.

(A3) **Elasticity.** Collisions conserve momentum and kinetic energy,

$$v + v_* = v' + v'_*, \qquad |v|^2 + |v_*|^2 = |v'|^2 + |v'_*|^2. \tag{A.5}$$

To exploit these hypotheses it is convenient to reformulate the two-sphere collision as an equivalent problem involving a *fixed sphere and a moving point*. Since two spheres of radius r are in contact exactly when their centers are separated by $\sigma = 2r$, this contact condition is unaffected if one sphere is inflated by r while the other is shrunk to a point. Concretely, we pass to the reference frame centered on the tagged particle and regard it as a *fixed* sphere of radius σ located at x, while the partner particle is idealized as a *point*, free to move around it; a collision occurs precisely when this point reaches the surface of the fixed sphere.

Let $\omega \in S^2$ be the unit vector, at the instant of contact, pointing from the center of the fixed sphere at x to the point of contact, equivalently, to the position of the incoming particle at that instant, so that

$$x_* = x + \sigma\,\omega, \qquad \omega = \frac{x_* - x}{\sigma} \in S^2. \tag{A.6}$$

An element of the collision surface, of area $\mathrm{d}S$, subtended by the solid angle $\mathrm{d}\omega$ around ω, is then

$$\mathrm{d}S = \sigma^2 \, \mathrm{d}\omega. \tag{A.7}$$

Since the two particles have equal mass, hypothesis (A3) forces the collision to act as a **specular reflection** of the relative velocity about the tangent plane at the point of contact: the component of the relative velocity along the line of centers ω is reversed, while the tangential component is left unchanged. Writing $V := v - v_*$ for the relative velocity just before impact, this gives the classical transformation rules

$$v' = v - (V \cdot \omega)\, \omega, \qquad v'_* = v_* + (V \cdot \omega)\, \omega. \tag{A.8}$$

It is straightforward to check that (A.8) indeed satisfies (A.5), i.e. $v' + v'_* = v + v_*$ and $|v'|^2 + |v'_*|^2 = |v|^2 + |v_*|^2$.

The pair $(\omega, \mathrm{d}S) = (\omega, \sigma^2 \mathrm{d}\omega)$ introduced here fully parametrizes the geometry of a single collision, and provides everything needed to turn the microscopic scattering picture into the gain and loss integrals, which we now compute.

A.4. Counting Collisions: The Two-Particle Distribution

To make the gain and loss terms explicit, we introduce the two-particle joint probability density $P^{(2)}(t, x, x_*, v, v_*)$: the likelihood of finding the tagged particle at (x, v) and a second particle at (x_*, v_*) at time t.

Because all N particles are identical and exchangeable, the tagged particle interacts symmetrically with each of the remaining $N-1$ particles. Consequently, the overall loss term factors as $L = (N-1)\ell$, where ℓ denotes the expected collision rate between the tagged particle and a single partner.

To compute ℓ we use the frame introduced above, in which the tagged particle at rest is a fixed sphere of radius σ: a partner with relative velocity $v_* - v$ strikes the surface element $\sigma^2 \, \mathrm{d}\omega$ within a time interval $\mathrm{d}t$ if and only if, at time t, it lies inside the slanted cylinder of base $\sigma^2 \, \mathrm{d}\omega$ and height $|(v_* - v) \cdot \omega| \, \mathrm{d}t$, swept out by that surface element along the incoming relative trajectory. Weighting this elementary volume by the density of partners at the corresponding phase-space point, and integrating over all relative velocities v_* and over the hemisphere of *approaching* directions,

$$B^- := \{\omega \in S^2 : (v_* - v) \cdot \omega < 0\}, \tag{A.9}$$

yields the loss term

$$L = (N-1)\, \sigma^2 \int_{\mathbb{R}^3} \int_{B^-} P^{(2)}(t, x, x + \sigma\omega, v, v_*)\, |(v_* - v) \cdot \omega| \, \mathrm{d}\omega \, \mathrm{d}v_*. \tag{A.10}$$

The gain term G follows from the symmetric argument, integrated instead over the hemisphere of *receding* directions,

$$B^+ := \{\omega \in S^2 : (v_* - v) \cdot \omega > 0\}, \tag{A.11}$$

which collects those pairs that have just separated after a collision producing the outgoing velocity v. Writing G directly as an integral over B^+ would require the density of partners *after* the collision has occurred; to bring it back to the same footing as L, one further hypothesis is needed.

(A4) **Continuity of $P^{(2)}$ across a collision.**

$$P^{(2)}(t, x, x_*, v, v_*) = P^{(2)}(t, x, x_*, v', v'_*). \tag{A.12}$$

Thus, the particle density in phase space changes smoothly and continuously over time and space, without sudden jumps or infinite spikes, as long as external forces and collisions are well-behaved.

Applying (A.12) together with the change of variable $\omega \mapsto -\omega$ (which exchanges B^+ and B^-), the gain term can likewise be written as an integral over B^-,

$$G = (N-1)\,\sigma^2 \int_{\mathbb{R}^3} \int_{B^-} P^{(2)}(t, x, x + \sigma\omega, v', v'_*)\,|(v_* - v)\cdot\omega|\,\mathrm{d}\omega\,\mathrm{d}v_*. \tag{A.13}$$

Equations (A.10) and (A.13) express L and G purely in terms of the two-particle distribution $P^{(2)}$; closing them in terms of $P^{(1)}$ alone is the object of the next two sections.

A.5. The Boltzmann–Grad Limit

The full balance expressions (A.10)–(A.13) still depend explicitly on the parameters N and σ. To pass to the continuum kinetic description while retaining a finite mean free path, we take the asymptotic scaling limit:

$$N \to \infty, \qquad \sigma \to 0, \qquad \text{such that} \quad N\sigma^2 \to C \in (0, \infty), \tag{A.14}$$

known as the **Boltzmann–Grad limit**.

In this limit, two important simplifications occur simultaneously:

1. **Pointwise collisions:** The spatial separation between colliding centers, $x_* = x + \sigma\omega$, vanishes as $\sigma \to 0$. Assuming $P^{(2)}$ is continuously differentiable in space, a Taylor expansion yields
$$P^{(2)}(t, x, x + \sigma\omega, v, v_*) = P^{(2)}(t, x, x, v, v_*) + \mathcal{O}(\sigma),$$
rendering binary collisions strictly local in space at position x.
2. **Finite collision frequency:** The pre-factor $(N-1)\sigma^2 \approx N\sigma^2$ converges to the non-zero constant C, preserving a finite rate of interaction per unit volume.

Absorbing the constant C into the scaling of the distribution function (or setting $C = 1$ without loss of generality via appropriate non-dimensionalization), the loss and gain terms simplify to:

$$L(t, x, v) = C \int_{\mathbb{R}^3} \int_{B^-} P^{(2)}(t, x, x, v, v_*)\,|(v_* - v)\cdot\omega|\,\mathrm{d}\omega\,\mathrm{d}v_*, \tag{A.15}$$

$$G(t, x, v) = C \int_{\mathbb{R}^3} \int_{B^-} P^{(2)}(t, x, x, v', v'_*)\,|(v_* - v)\cdot\omega|\,\mathrm{d}\omega\,\mathrm{d}v_*. \tag{A.16}$$

A.6. The Molecular Chaos Assumption and Closure

Equations (A.15)–(A.16) are still not closed: they express the evolution of $P^{(1)}$ in terms of the unknown two-particle distribution $P^{(2)}$, about which nothing has been assumed so far. This is where Boltzmann's central hypothesis enters.

(A5) **Molecular chaos (*Stosszahlansatz*).** Because N is large, a collision between two specific, pre-selected particles is a rare event; encounters are treated as random pairings of statistically independent particles. Concretely, the velocities of two particles about to collide are assumed uncorrelated,

$$P^{(2)}(t, x, x_*, v, v_*) = P^{(1)}(t, x, v)\, P^{(1)}(t, x_*, v_*), \qquad (v_* - v) \cdot \omega < 0. \tag{A.17}$$

Substituting (A.17) into (A.15) expresses the loss term solely in terms of $P^{(1)}$. For the gain term (A.16), which depends on the post-collisional velocities (v', v'_*), we invoke the continuity hypothesis (A4) once more to evaluate the same factorized form at (v', v'_*). Adopting the standard shorthand $f = f(t, x, v)$, $f_* = f(t, x, v_*)$, $f' = f(t, x, v')$, and $f'_* = f(t, x, v'_*)$, and rescaling to absorb the constant C, we arrive at the closed, self-contained evolution equation

$$\partial_t f + v \cdot \nabla_x f = \int_{\mathbb{R}^3} \int_{B^-} \left[f' f'_* - f f_* \right] |(v_* - v) \cdot \omega| \, \mathrm{d}\omega \, \mathrm{d}v_*, \tag{A.18}$$

which is the **Boltzmann equation for hard spheres**.

In particular, hypothesis (A5) hide an important consequence. Strict classical (Newtonian) dynamics is time-reversible, and even if the pre-collisional velocities of two particles were exactly uncorrelated, the act of colliding necessarily correlates their post-collisional velocities; assumption (A5), applied simultaneously, via (A4), to pre- and post-collisional pairs, cannot hold exactly at the level of the deterministic N-body flow. It must be understood only as a statement that these correlations are negligible in the scaling limit (namely the Boltzmann–Grad limit (A.14)). In this step the derivation departs from reversible microscopic mechanics and introduces the time-irreversibility responsible for the H-theorem and the monotone decay of entropy shown in Chapter 2.

A.7. General Collision Kernels

The derivation above used the hard-sphere billiard geometry, the specular reflection law (A.8), to compute the weight $|(v_* - v) \cdot \omega|$ with which a collision produces a given outgoing direction ω. For a general short-range interaction potential, the elastic conservation laws (A.5) still force the post-collisional velocities onto the same sphere as before, but the relation between the impact geometry and the resulting scattering direction and hence the relative likelihood of each outgoing ω is instead dictated by the physics of the potential (e.g. a Coulomb or an inverse-power force law), rather than by simple specular reflection. Formally, this amounts to replacing the hard-sphere weight $|(v_* - v) \cdot \omega|$ by a general nonnegative *collision kernel* $\sigma(\omega, |v - v_*|)$, and extending the integration from the hemisphere B^- to the full sphere S^2 a standard reformulation that folds the incoming/outgoing

labeling into the kernel itself. This yields the general Boltzmann equation

$$\partial_t f + v \cdot \nabla_x f = \int_{\mathbb{R}^3} \int_{S^2} \left[f' f'_* - f f_* \right] \sigma(\omega, |v - v_*|) \, \mathrm{d}\omega \, \mathrm{d}v_* \; = \; Q(f, f), \tag{A.19}$$

All of the physics of the interaction is encoded in σ.

A.8. The Knudsen Number and the Scaled Boltzmann Equation

The closed equation (A.19) is naturally expressed in *kinetic* units: space and time have implicitly been measured on the scale of the mean free path and of the mean time between collisions, precisely the scale singled out by the Boltzmann–Grad limit. To connect this equation to the form used throughout the main text, namely equation (2.1) of Chapter 2, we must re-express it in *macroscopic* units, appropriate to an observer working at the scale of the container or obstacle rather than at the scale of a single molecule.

Let λ denote the mean free path of the gas: the average distance a molecule travels between two successive collisions, and let L be a characteristic macroscopic length of the problem (e.g. the size of the domain Ω). Their ratio defines the **Knudsen number**,

$$\varepsilon := \frac{\lambda}{L}. \tag{A.20}$$

The Knudsen number measures how rarefied the gas is *relative to the problem at hand*: small ε means the mean free path is negligible compared to L, i.e. a dense, near-continuum gas, while large ε means molecules typically travel a macroscopic distance before colliding, i.e. a strongly rarefied gas. Table A.1 lists the flow regimes conventionally associated with different ranges of ε.

Regime	Range of ε
Continuum flow	$\varepsilon \leq 10^{-3}$
Slip flow	$10^{-3} < \varepsilon \leq 10^{-1}$
Transition flow	$10^{-1} < \varepsilon \leq 10$
Free molecular flow	$\varepsilon > 10$

Table A.1: Flow regimes classified by the Knudsen number.

To see explicitly where ε enters the equation, we define a characteristic macroscopic velocity U (e.g. the thermal speed of the gas) and the associated macroscopic time scale $T := L/U$, and rescale

$$\tilde{x} := \frac{x}{L}, \qquad \tilde{v} := \frac{v}{U}, \qquad \tilde{t} := \frac{t}{T} = \frac{tU}{L}. \tag{A.21}$$

By the chain rule, the free-streaming part of (A.19) scales uniformly by U/L,

$$\partial_t f = \frac{U}{L} \, \partial_{\tilde{t}} \tilde{f}, \qquad v \cdot \nabla_x f = \frac{U}{L} \, \tilde{v} \cdot \nabla_{\tilde{x}} \tilde{f}. \tag{A.22}$$

The collision operator $Q(f, f)$, on the other hand, has the dimensions of an inverse time, specifically, the inverse of the mean time between collisions, $\tau := \lambda/U$, so that $Q(f, f) = (U/\lambda)\,\tilde{Q}(\tilde{f}, \tilde{f})$, with $\tilde{Q}$ a dimensionless operator of the same functional form as Q. Substituting both scalings into (A.19) and dividing through by U/L gives

$$\partial_{\tilde{t}}\tilde{f} + \tilde{v} \cdot \nabla_{\tilde{x}}\tilde{f} = \frac{L}{\lambda}\,\tilde{Q}(\tilde{f}, \tilde{f}) = \frac{1}{\varepsilon}\,\tilde{Q}(\tilde{f}, \tilde{f}). \tag{A.23}$$

Dropping the tildes, as is conventional once the rescaling has been performed, we recover precisely the scaled Boltzmann equation used throughout the main text,

$$\partial_t f + v \cdot \nabla_x f = \frac{1}{\varepsilon}\,Q(f, f), \tag{A.24}$$

i.e. equation (2.1) of Chapter 2.

In this equation it is possible to understand that as $\varepsilon \to 0$, the factor $1/\varepsilon$ forces $Q(f, f) \to 0$ on any time scale of order one, so that f relaxes to a local Maxwellian and the fluid-dynamic (Euler) description becomes valid; as $\varepsilon \to \infty$, the collision term becomes negligible compared to free streaming, and the dynamics reduce to free molecular flow. It is precisely ε that controls the stiffness of the collision operator responsible for the numerical difficulties discussed in the main text.

B | Appendix B

Section 3.5 introduced the two basic ideas: inversion of the cumulative distribution function, and acceptance-rejection against a majorant, that allow one to turn a source of independent uniform random numbers $\xi \sim \mathcal{U}[0,1]$ into a sample from a prescribed probability density p_x. Every Monte Carlo particle method for the Boltzmann equation relies, at some point, on this operation: the DSMC and TRMC schemes of Chapther 4 need it to build the initial particle system from f_0, to draw a post-collisional direction $\omega \in S^2$ from the scattering kernel $\sigma(\omega, |v - v_*|)$ once a colliding pair has been selected, and to re-thermalise particles hitting a wall according to the Maxwellian re-emission law (2.8). This appendix collects, in a self-contained way and following the presentation of [3], the general-purpose sampling algorithms needed throughout the report: for scalar and discrete random variables in Section B.1, and for multivariate random variables in Section B.2, together with the two examples of direct relevance to kinetic theory: the Gaussian (hence Maxwellian) and the uniform distribution on the sphere.

Throughout, $\xi, \xi_1, \xi_2, \ldots$ denote independent random numbers, uniformly distributed on $[0,1]$, drawn at every occurrence.

B.1. Univariate and discrete distributions

B.1.1. Inversion

Let x be a real random variable with density $p_x(x) \geq 0$, $\int_{\mathbb{R}} p_x(x)\,dx = 1$, and cumulative distribution function $P_x(x) = \int_{-\infty}^{x} p_x(y)\,dy$. Equating the infinitesimal probabilities $p_x(x)\,dx = d\xi$ and integrating gives $P_x(x) = \xi$, so that x can be produced from a single uniform random number by solving this equation for x:

Algorithm B.1 (Inversion).

1. draw $\xi \sim \mathcal{U}[0,1]$;
2. set $x = P_x^{-1}(\xi)$.

This is exactly (3.12), restated here for completeness. The method is exact and requires a single random number per sample, but it can only be used when P_x can be inverted, in closed form or numerically at negligible cost.

B.1.2. Acceptance-rejection

When P_x^{-1} is unavailable or too expensive, inversion is replaced by acceptance-rejection, already introduced in Section 3.5 and recalled here for reference. Let $w(x) \geq p_x(x)$ for every x, with w chosen so that it is itself easy to sample (e.g. by inversion of its primitive W), and let $A = \int w(x)\,dx$.

Algorithm B.2 (Acceptance-rejection).

1. sample a candidate x from the density $w(x)/A$;
2. draw $\xi \sim \mathcal{U}[0,1]$ independently; if $w(x)\,\xi < p_x(x)$, accept x; otherwise reject it and return to step 1.

As shown in Section 3.5, the accepted samples are distributed exactly according to p_x, and the average fraction of accepted candidates is $1/A$: the closer the majorant w hugs p_x, the more efficient the algorithm. Figure B.1 illustrates the geometric content of the method: a candidate η, drawn from w, is accepted with probability $p_x(\eta)/w(\eta)$, i.e. it is accepted if and only if it also lies, in the vertical direction, under the graph of p_x.

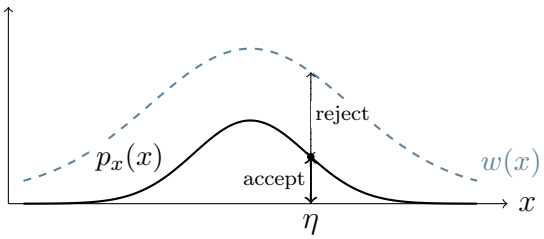


Figure B.1: Acceptance–rejection: a candidate η is drawn from the majorant density $w(x)/A$; it is accepted with probability $p_x(\eta)/w(\eta)$, i.e., when it lies, in the vertical direction, under the graph of p_x.

B.1.3. Composition

Frequently a density is most naturally given as a convex combination of simpler densities,

$$p_x(x) = \sum_{i=1}^{M} w_i\, p_i(x), \qquad w_i \geq 0,\ \sum_{i=1}^{M} w_i = 1,$$

each p_i being easy to sample individually (by inversion or acceptance-rejection). Sampling p_x then reduces to first choosing which component to sample from, with the corresponding probability, and then sampling that component:

Algorithm B.3 (Composition).

1. select an index $i \in \{1, \ldots, M\}$ with probability w_i (see Section B.1.4 below);
2. sample x from the density $p_i(x)$.

This is elementary but useful: it is, for instance, the natural way to sample from a scattering kernel that is itself a mixture of an isotropic part and a forward-peaked part, or from a bimodal initial datum f_0.

B.1.4. Discrete distributions

Step 1 of Algorithm B.3 requires sampling an integer $k \in \{1, \dots, M\}$ with prescribed probability w_k. Two situations occur in practice.

The first is when the weights $\{w_k\}$ are known and fixed. The interval $[0, 1]$ is partitioned into M consecutive sub-intervals of length $w_1, \dots, w_M$, and k is the index of the sub-interval containing a uniform random number:

Algorithm B.4 (Discrete inversion).

1. compute the cumulative weights $W_k = \sum_{i=1}^{k} w_i$, $k = 1, \dots, M$, with $W_0 = 0$;
2. draw $\xi \sim \mathcal{U}[0, 1]$ and find the (unique) index k such that $W_{k-1} \le \xi < W_k$.

Once the W_k have been precomputed, step 2 can be carried out by binary search in $O(\log M)$ operations, which is efficient whenever the same table of weights is reused many times.

Instead if the weights $\{w_k\}$ are numerous, or change at every sampling. Precomputing and searching the cumulative table is then wasteful. If a uniform bound $\overline{w} \ge w_k$ for $k = 1, \dots, M$ is available (rather than the w_k themselves), acceptance-rejection on the index k can be used instead:

Algorithm B.5 (Discrete acceptance-rejection).

1. draw $\xi_1 \sim \mathcal{U}[0, 1]$ and set $k = \lfloor M\xi_1 \rfloor + 1$, a uniformly distributed integer in $\{1, \dots, M\}$ (be aware of the floor function of $M\xi_1$, $\lfloor M\xi_1 \rfloor$);
2. draw $\xi_2 \sim \mathcal{U}[0, 1]$; if $w_k\, \xi_2 < \overline{w}$, accept k; otherwise reject it and return to step 1.

This variant is used to sample a candidate collision partner (or a candidate pair) uniformly at random, then accept it with a probability governed by the actual transition rate is exactly the mechanism used by Bird's and Nanbu's schemes to turn the collision kernel σ into a selection rule for colliding pairs, and reappears in Section 4.1 in that guise.

B.1.5. Sampling without repetition

A related, purely combinatorial, need arises when a subset of $n \le N$ *distinct* indices must be drawn out of $\{1, \dots, N\}$, without repetition and without bias, for instance, when the N_p simulation particles must be split into $N_p/2$ disjoint colliding pairs at each time step. An efficient way to do this is the following swap-to-the-end procedure.

Algorithm B.6 (Sampling without repetition).

1. initialize $\texttt{ind}_i = i$ for $i = 1, \dots, N$, and set $M = N$;

2. for $i = 1, \dots, n$: draw $\xi \sim \mathcal{U}[0,1]$, set $j = \lfloor M\xi \rfloor + 1$, output $\mathtt{seq}_i = \mathtt{ind}_j$, then overwrite $\mathtt{ind}_j = \mathtt{ind}_M$ and set $M = M - 1$.

At every iteration the active pool $\{\mathtt{ind}_1, \dots, \mathtt{ind}_M\}$ consists precisely of the indices not yet selected, and step i selects one of them uniformly at random (since j is uniform on $\{1, \dots, M\}$); overwriting the chosen slot with the last active one removes it from the pool in $O(1)$ operations without disturbing the uniformity of the remaining choices. By induction, the output $\mathtt{seq}_1, \dots, \mathtt{seq}_n$ is therefore a uniformly random ordered sample of n distinct elements of $\{1, \dots, N\}$; for $n = N$ it produces a uniformly random permutation.

B.2. Multivariate distributions

B.2.1. The factorised case

Suppose we must sample an n-dimensional random variable $x = (x_1, \dots, x_n)$ with density $p_x(x_1, \dots, x_n)$. If the density factorises into a product of one-dimensional marginals,

$$p_x(x_1, \dots, x_n) = p_1(x_1)\, p_2(x_2) \cdots p_n(x_n), \tag{B.1}$$

then $x_1, \dots, x_n$ are independent and each component can be sampled separately by any of the univariate methods of Section B.1. This is the situation that occurs, for instance, for a Maxwellian velocity distribution, whose three velocity components are independent Gaussians (see Example B.2 below).

B.2.2. The general case: sequential conditioning

If (B.1) does not hold, one looks for a change of variables $\eta = T(x)$ under which the transformed density factorises,

$$p_x(x_1, \dots, x_n)\, dx_1 \cdots dx_n = p_{\eta_1}(\eta_1)\, p_{\eta_2}(\eta_2) \cdots p_{\eta_n}(\eta_n)\, d\eta_1 \cdots d\eta_n,$$

so that the η_i can be sampled independently and x recovered as $x = T^{-1}(\eta)$. Such a T always exists (it is far from unique, since only its Jacobian $|\partial\eta/\partial x| = p_x$ is prescribed) and can be constructed explicitly as the map to the successive *conditional* cumulative distribution functions of $x_1, \dots, x_n$:

$$\begin{aligned}
T_1(x_1) &= \int_{-\infty}^{x_1} d\eta \int_{\mathbb{R}^{n-1}} dx_2 \cdots dx_n \; p_x(\eta, x_2, \dots, x_n), \\
T_2(x_1, x_2) &= \frac{\displaystyle\int_{-\infty}^{x_2} d\eta \int_{\mathbb{R}^{n-2}} dx_3 \cdots dx_n \; p_x(x_1, \eta, x_3, \dots, x_n)}{\displaystyle\int_{-\infty}^{\infty} d\eta \int_{\mathbb{R}^{n-2}} dx_3 \cdots dx_n \; p_x(x_1, \eta, x_3, \dots, x_n)}, \\
&\;\;\vdots \\
T_n(x_1, \dots, x_n) &= \frac{\displaystyle\int_{-\infty}^{x_n} d\eta \; p_x(x_1, \dots, x_{n-1}, \eta)}{\displaystyle\int_{-\infty}^{\infty} d\eta \; p_x(x_1, \dots, x_{n-1}, \eta)},
\end{aligned} \tag{B.2}$$

i.e. $T_k(x_1,\dots,x_k)$ is the cumulative distribution function of x_k conditioned on $x_1,\dots,x_{k-1}$. One checks directly that $|\partial\eta/\partial x| = p_x(x_1,\dots,x_n)$, so that $\eta = T(x)$ is uniform on $[0,1]^n$ whenever $x \sim p_x$; conversely, given $\eta \sim \mathcal{U}[0,1]^n$, $x = T^{-1}(\eta)$ is recovered one coordinate at a time by solving, in order, $T_1(x_1) = \eta_1$ for x_1, then $T_2(x_1,x_2) = \eta_2$ for x_2 (with x_1 already known) and so on, thus the system (B.2) is triangular, and its inversion occurs as n successive *scalar* inversions rather than one n-dimensional problem.

B.2.3. Multivariate acceptance-rejection

When the conditional distribution functions (B.2) are too costly to compute, the acceptance-rejection method of Section B.1.2 generalizes to $\mathbb{R}^n$: given a majorant $w(x) \geq p_x(x)$ on $\mathbb{R}^n$, easy to sample, with $A = \int_{\mathbb{R}^n} w(x)\,dx$,

Algorithm B.7 (Multivariate acceptance-rejection).

1. sample a candidate $x \in \mathbb{R}^n$ from the density $w(x)/A$, by any of the methods above (e.g. coordinate-wise, if w factorises);
2. draw $\xi \sim \mathcal{U}[0,1]$; if $w(x)\,\xi < p_x(x)$, accept x; otherwise reject it and return to step 1.

This is the method of choice whenever a scattering kernel $\sigma(\omega)$ in Section 4.1 is anisotropic: one samples a candidate direction ω uniformly on S^2 (Example B.2 below), and accepts it with probability proportional to $\sigma(\omega)$ relative to its maximum.

Example B.1: sampling a Gaussian (Box-Muller method)

The standard Gaussian density $p(x) = (2\pi)^{-1/2}e^{-x^2/2}$ has cumulative distribution function $P(x) = \frac{1}{2}\big(1+\operatorname{erf}(x/\sqrt{2})\big)$, whose inversion requires the (expensive) inverse error function. The *Box-Muller method* avoids this by sampling *two* independent standard Gaussians x, y at once, using their joint density in polar coordinates $x = \rho\cos\theta$, $y = \rho\sin\theta$:

$$\frac{1}{2\pi}\exp\!\left(-\frac{x^2+y^2}{2}\right)dx\,dy = \Big[\underbrace{\rho\,e^{-\rho^2/2}}_{p_\rho(\rho),\ \rho\geq 0}\,d\rho\Big]\Big[\underbrace{\frac{1}{2\pi}}_{p_\theta(\theta),\ 0\leq\theta<2\pi}\,d\theta\Big],$$

a product of two univariate densities, each trivially invertible:

$$\rho = \sqrt{-2\ln\xi_1}, \qquad \theta = 2\pi\xi_2, \qquad x = \rho\cos\theta, \quad y = \rho\sin\theta. \tag{B.3}$$

Thus, from ξ_1 and ξ_2 we can sample x and y with (B.3) distributed as the standard Gaussian, if instead we consider mean μ and standard deviation σ the sampling has to be carried out considering $\mu + \sigma x$.

This is directly relevant to the Boltzmann equation because the local Maxwellian (2.13),

$$M(\rho,u,T)(v) = \frac{\rho}{(2\pi T)^{3/2}}\exp\!\left(-\frac{|v-u|^2}{2T}\right), \qquad v = (v_1,v_2,v_3) \in \mathbb{R}^3,$$

factorises, as in (B.1), into three independent Gaussian components, each of mean u_i and variance T. A velocity sampled from M is thus obtained by two applications of (B.3): the first supplies v_1, v_2 (with $\mu = u_1, u_2$ and $\sigma = \sqrt{T}$), and the second supplies v_3 (using only one of the two Gaussians it produces, or reusing the 4th spare one for the next particle). This is the operation needed to initialise a DSMC/TRMC particle system from a Maxwellian initial datum f_0, and to re-emit particles from the wall according to the thermalisation operator (2.9).

Example B.2: sampling a uniform direction on the sphere S^2

A unit vector $\omega \in S^2$ is described by two angles (φ, θ),

$$\omega = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta), \qquad \varphi \in [0, 2\pi),\ \theta \in [0, \pi],$$

and ω is uniformly distributed on the sphere precisely when the probability of falling in a given region is proportional to the solid angle it subtends,

$$dP = \frac{d\Omega}{4\pi} = \frac{\sin\theta\, d\theta}{2} \cdot \frac{d\varphi}{2\pi}.$$

The two factors on the right are again univariate densities (φ uniform on $[0, 2\pi)$, and θ with density $(\sin\theta)/2$ on $[0, \pi]$), they can be inverted by Algorithm B.1 as:

$$\varphi = 2\pi\xi_1, \qquad \theta = \arccos(1 - 2\xi_2). \tag{B.4}$$

Formula (B.4) is exactly what is needed to implement the ω-representation (2.4) of the post-collisional velocities in the isotropic case, the Kac model and Maxwell molecules with constant angular kernel, where ω is, by definition, uniformly distributed on S^2 and no acceptance-rejection is required. For a general, angle-dependent kernel $\sigma(\omega, |v - v_*|)$, ω is instead sampled by drawing a uniform candidate through (B.4) and applying Algorithm B.7 (or its scalar counterpart) with σ itself, suitably normalised, playing the role of p_x and the uniform density playing the role of the majorant w.

List of Figures

4.1 Binary collision trees (genealogies) for generations f_1– f_3. Internal white nodes denote binary collision events $P(\cdot, \cdot)$, and shaded leaf nodes denote initial uncollided particles drawn from f_0. Every generation f_n requires n internal collision nodes and exactly $n + 1$ original leaf particles, regardless of whether the tree topology is asymmetric (as in c) or symmetric (as in d). 27
4.2 Generation of particles via binary collision tree in the TRMC algorithm. . 28

B.1 Acceptance–rejection: a candidate η is drawn from the majorant density $w(x)/A$; it is accepted with probability $p_x(\eta)/w(\eta)$, i.e., when it lies, in the vertical direction, under the graph of p_x. 44

List of Tables

A.1 Flow regimes classified by the Knudsen number. 40